WILL

Relational Geometry

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Abstract

This paper, the first in the **WILL Trilogy:**
Part I: Relational Geometry;
Part II: Relational Cosmology;
Part III: Relational Quantum Mechanics (R.Q.M.).
The main goal of this Open Research is to derive the laws of physical reality without introducing new postulates, axioms, or arbitrary choices. Laws of nature should not rely on pre-existing formalism but naturally arise from the self consistency and geometry. By applying extreme methodological constraints we established *Relational Geometry (RG)*: a background free relational foundation.

The transition from *descriptive* to *generative* physics: instead of introducing laws to model observations, we derive them as necessary consequences of methodological principles - turning physics from a catalogue of phenomena into the logical unfolding of inevitable geometrical constraints.

Without metrics, tensors, or free parameters, we reproduce Lorentz factors, the energy-momentum relation, Minkowski and Schwarzschild metric intervals and Einstein field equations via the dimensionless projections of state difference β (kinematic) and κ (potential). All known GR critical surfaces (photon sphere, ISCO, horizons) emerge as simple fractions of (κ, β).

***This paper* offers new potential solutions to several long-standing problems, including:**

- **Derivation of Relational Orbital Mechanics** (Background-free and algebraically closed system of equation for orbital systems),
- **Resolution of GR singularities** (via naturally bounded radial distance $r_{min} = R_s/2$ and density $\rho_{\max} = \frac{c^2}{8\pi G r^2}$ (4.9)),
- **Derivation of the equivalence principle** (from the common channel of rest-invariant state scaling (4.3, 4.2),
- **Removal of local energy density ambiguity** $\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$ (4.6.1),
- **Revelation of a clear relational symmetry between kinematic and potential** energy's (3.9),
- **Establishment of a computationally simpler and ontologically consistent foundation** (for subsequent papers on cosmology (Part II) and quantum mechanics (Part III)).

A Direct Note to the Reader: Why should you read a 30+ page independent open research paper?

The fact that you opened this file and reading these lines already separates you from the majority, and for that, you have my respect. However, 30+ pages of unknown research is a substantial commitment, and you should not engage it blindly.

I recommend the following: Be honest with yourself. Define the exact criteria this paper must satisfy to justify the investment of your time. Then, ask WILL-AI if this work meets your standards. The AI is trained on this research documents and will give you an honest, unbiased answer based on its contents.

If you are still in doubt, browse willrg.com, interact with the Galactic Dynamics Lab, trace the derivation chain using the Logos Map, review the falsifiable predictions, and have a chat or few with WILL-AI .

Regardless of your decision, I wish you to find what you are seeking.

Anton Rize

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Following yet untamed thought

"There is no such thing as an empty space, i.e., a space without field. . . . Space-time does not claim existence on its own, but only as a structural quality of the field."
— *Albert Einstein, Relativity: The Special and the General Theory* (Appendix V: "Relativity and the Problem of Space"), 1952 edition, Methuen (London), p. 155; based on earlier 1920 additions.

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IMPORTANT:

This document must be read **literally**. All terms are defined within the relational logic of WILL Relational Geometry.
Definition of Geometry: The term "geometry" is used herein in its fundamental, classical sense: the pure logic of relations, ratios, and structural constructions. It **does not** refer to modern differential geometry. There is no *a priori* metric tensor ($g_{\mu\nu}$), no Riemannian manifold, and no underlying coordinate grid. The geometry emerges from physical relations, not analytically from a background space.
Any attempt to reinterpret these derivations through conventional metric notions (*absolute distances, external backgrounds, hidden containers*) will produce distortions and mathematical misreadings.
Take the words as written and ontology as derived, without smuggling external concepts.

1 Stage 0: The Last Geocentric Epicycle

The standard formulation of General Relativity relies on the concept of a pre-existing, container-like spacetime that then gets "filled" with fields and matter.

Derivation of the Einstein field equations begins with the Einstein-Hilbert action, which is built upon the metric as the fundamental variable. This metric is defined on a smooth manifold that is assumed to exist *a priori*. This manifold, even without a metric, carries topological and differential structure - an absolute scaffold. This constitutes surplus ontology because it introduces an entity (the manifold) that is not derived neither observed.

The stress-energy tensor is derived from the variation of the matter Lagrangian with respect to the metric. This assumes that energy is a property of matter fields that can be localized in spacetime. However, this localization is frame-dependent (via the equivalence principle) and leads to well-known problems such as the non-uniqueness of the gravitational energy-momentum pseudotensor. Energy is treated as an absolute property of matter rather than a relational measure. This remains as metaphysical speculation in the heart of modern physics.

1.1 The contemporary split: an unjustified ontological commitment

All present-day theories (SR, GR, QFT, CDM, Standard Model) are built with a *bi-variable* syntax:

$$\underbrace{\text{fixed manifold} + \text{metric}}_{\text{structure}} \quad + \quad \underbrace{\text{fields} + \text{constants}}_{\text{dynamics}}.$$

No observation demands this duplication; it is retained purely because the resulting Lagrangians are empirically adequate *inside* the split. The split is therefore *not* an empirical discovery but an unjustified ontological statement.

1.2 Empirical deficit of the separation

- **Local energy conservation** verified only *after* the metric is declared fixed; no experiment varies the *volume* of flat space and checks calorimetry.
- **Universality of free fall** tests $m_i = m_g$ numerically, not the claim that inertia resides *in* the object rather than in a geometric scaling relation.
- **Gravitational-wave polarisations** test spin content, not ontology; extra modes can still be called "matter on spacetime".
- **Casimir/Lamb shifts** measure *differences* of vacuum energy between two geometries; the absolute bulk term is explicitly subtracted, leaving the split intact.

Every "test" is an *internal consistency check* of a formalism that already presupposes two substances. None constitute *positive evidence* for the split.

1.3 Historical Pattern: breakthroughs delete, not add

- **Copernicus** eliminated the Earth/cosmos separation.
- **Newton** eliminated the terrestrial/celestial law separation.
- **Maxwell** eliminated the electricity/magnetism separation.
- **Einstein** eliminated the space/time separation.

Each step widened the relational circle and reduced the number of primitives - an unexplained absolutes. The spacetime–energy split is the only survivor of this pruning sequence.

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1.4 Consequence

Until an experiment varies the amount of space while holding everything else fixed, the spacetime–energy separation remains an *unevidenced metaphysical postulate.* **It is the last geocentric epicycle in physics.**

2 Stage I: Ontological Construction (The Primitives)

2.1 Foundational Methodological Principles

> *Every logical step is subject to empirical audit: if any consequence of this chain contradicts observation, the chain fails.*

As humans, we often project ourselves onto observed phenomena, obscuring the true nature of reality behind our anthropic tendencies. In order to prevent such misleading practices, we must establish extreme methodological constraints. Not as statements about the nature of reality, but strict rules of logical and epistemic purity. Without such methodology, any theoretical construct risks that excessive freedom will result in a framework that tells us more about ourselves than about the subject at study.

Primary Goal:

Derive the laws of physical reality from the most elementary principles by peeling-off interpretational layers until the foundational primitives revealed.

Traditionally a new theory is expected to propose new undetected fields or particles, postulates or axioms - a priori statements about physical reality. This research In contrast, adheres strictly to methodological constraints. The aim is for the theory to arise naturally from the geometry and logic itself. The foundational core of Relational Geometry - its principles and theorems - emerge from methodological constraints. They function as inescapable outcomes of the method itself.

The philosophical reasoning behind these constraints probably deserves its own dedicated publication and can be challenged. However, this research's main domain lies within physical reality that can be empirically tested; therefore, empirical alignment remains the ultimate test of the methods and the derived results they produce.

Flow chart of derivation logic

Logos Map

Principle 2.1 (Epistemic Hygiene)**.** Epistemic Hygiene as Refusal to Import Unjustified Assumptions. *This line of reasoning derive physics by* ***removing hidden assumptions****, rather than introducing new postulates. This construction is deliberate and contains zero free parameters. This is not a simplification - it is a deliberate epistemic constraint. No assumptions are introduced and no constructs are retained unless they are geometrically or energetically necessary.*

Principle 2.2 (Relational Origin)**.** All physical quantities must be defined by their relations. *Any introduction of absolute properties risks reintroducing metaphysical artefacts and contradicts the foundational insight of relationalism.*

Principle 2.3 (Ontological Minimalism)**.** *Maintain the minimum number of ontological assumptions and foundational primitives. Absolute background — spatial container, axis of time, absolute distance, forces, fields — are excluded as preexisting primitives but may appear as translation into common descriptive vocabulary.*

Principle 2.4 (Mathematical Transparency)**.**

1. *Every mathematical phrase, operational choice, or identity carries its own ontological statement.*
2. *Each mathematical object must correspond to explicitly identifiable relation between observers with transparent ontological origin.*
3. *Every symbol must be anchored to unique physical idea.*
4. *Introducing symbols without explicit necessity constitutes* ***Over-parameterization****: the proliferation of symbols without corresponding physical meaning.*
5. *Number of symbols = Number of independent physical ideas.*

> *Mathematics is a language, not a world. Its symbols must never outnumber the physical meanings they encode.*

2.1.1 The Foundational Core: Relational Geometry

This foundational methodological principles lead us to the core of Relational Geometry:

Theorem 2.5 (Relational Closure)**.** *A purely relational system cannot be geometrically open; Any stable, self-contained system must be described by a closed and self-consistent relational geometry. Its geometry is closed by absolute analytic necessity.*

Proof by Analytic Tautology.

1. **Premise 1 (Relational Origin):** Within RG Physical quantities are defined exclusively by their relations (Principle 2.2).
2. **Premise 2 (Interaction as Integration):** If a hypothetical "outside" entity or boundary were to physically interact with the system, that interaction itself constitutes a relation.
3. **Deduction (The Tautology):** As relation is formed, it is mathematically and physically integrated into the system.

An "open relational system" experiencing external influence is a logical contradiction. The system is closed not by a spatial boundary or a metric container, but by the strict impossibility of an interacting non-relation. The system is relationally closed by definition. □

2.1.2 What is Energy in Relational Ontology?

> Across all domains of physics, one empirical fact persists: in every closed system there exists a quantity that never disappears or arises spontaneously, but only transforms in form. This invariant is observed under many guises — kinetic, potential, thermal, chemical — yet all are interchangeable, pointing to a single underlying structure.
>
> Crucially, this quantity is never observed directly, but only through *differences between states*: a change of velocity, a shift in configuration, a transition of phase. Its value is relational, not absolute: it depends on the chosen frame or comparison, never on an object in isolation.
>
> Moreover, this quantity provides continuity of causality. If it changes in one part of the system, a complementary change must occur elsewhere, ensuring the unbroken chain of transitions. Thus it is the bookkeeping of causality itself.

Theorem 2.6 (Relational Invariance)**.** *Within relationally closed system without an external background, the existence of change necessitates the existence of conserved invariant measure of change. Because the system is closed any change must be perfectly balanced by a complementary change elsewhere in the system conserving the total transformations.*

Proof by Structural Necessity.

1. **Premise 1 (Relational Origin):** Physical quantities are defined exclusively by their relations (Principle 2.2).
2. **Premise 2 (Empirical Change):** States undergo observable change.
3. **Premise 3 (Relational Closure):** As proven in Theorem 2.5, the relational system is analytically closed. There is no external background, void, or hidden container to absorb or inject influence.
4. **Deduction (Causal Continuity):** Because the system is closed (Premise 3), an empirical change (Premise 2) cannot vanish into a void or emerge from one. To maintain the structural integrity of the relations (Premise 1), any change must be perfectly balanced by a complementary change elsewhere in the system.

Therefore, within this closed relational network, there must exist an invariant quantitative measure of change. □

Corollary 2.7 (Historical Note on Energy)**.** *In classical and relativistic mechanics, this derived invariant measure of change is assigned the historical label "Energy". By proving this invariance geometrically, we strip "Energy" of its substantialist properties (as a "substance" or "thing") and reveal its origin as the necessary measure of change and bookkeeping of causality.*

Energy :

Definition 2.8 (Energy)**.**

Energy is the invariant relational measure of state transformation within relationally closed system.

Theorem 2.9 (Relational Isotropy)**.** *Within a background-free relational system, no spatial direction or coordinate can be* a priori *privileged. Therefore, the geometry encoding the conserved relational resource (energy) must be maximally symmetric (isotropic).*

Proof by Relational Origin.

1. **Premise 1 (Absence of a Grid):** As established by Theorem 2.5 (Relational Closure), the relational system possesses no external container, absolute metric tensor, or underlying coordinate grid.

2. **Premise 2 (Direction requires Relation):** Under the Principle of Relational Origin (Principle 2.2), "direction" or "orientation" cannot exist in a void; it only physically manifests as a specific interaction or measurement between defined participants.

3. **Deduction (Pure Gauge):** Because there is no absolute grid (Premise 1), assigning an intrinsic, absolute axis to the unmeasured relational resource (energy) introduces unobservable "surplus ontology." Any such arbitrary orientation is a mathematical artifact (pure gauge).

Therefore, prior to a specific interaction defining a local axis, the pure relational structure encoding this conserved resource must possess no intrinsic privileged direction. It must be strictly isotropic. □

2.2 Unifying Ontological Principle

Summary

Any attempt to treat *"spacetime structure"* as separate from *"dynamics"* smuggles in a background container that is not justified by the phenomena. This violates epistemic hygiene: it introduces an ontological artifact without necessity. Eliminating this separation compels the identification of structure and dynamics as two aspects of a single entity.

Lemma 2.10 (False Separation)**.** *Any model that treats processes as unfolding* within *an independent background necessarily assigns to that background structural features (metric, orientation, or frame) not derivable from the relations among the processes themselves. Such a background constitutes an* extraneous absolute.

Proof. Suppose an independent background exists. Then at least one of its structural attributes - metric relations, a preferred orientation, or a class of inertial frames - remains fixed regardless of inter-process data. This attribute is not relationally inferred but posited a priori. It thereby violates the relational closure principle: it introduces a non-relational absolute external to the system. Hence the separation is illicit. □

Corollary 2.11 (Structure–Dynamics Coincidence)**.** *Without introducing new primitives 2.10, we should see structure and dynamics as a single ontological element. The structural arena and the dynamical content must be identified: structure is dynamics, energy is spacetime.*

This leads us to principle with *negative ontological weight*: it reduces the number of primitives and removes the separation between structure and dynamics. Spacetime is not a "container" but relational tension between energy states.

Principle 2.12 (Unifying Ontological Principle)**.**

$$\boxed{\textbf{\textit{SPACETIME}} \equiv \textbf{\textit{ENERGY}}}$$

No background container permitted.

Definition 2.13 (WILL)**. WILL ≡ SPACE-TIME-ENERGY** *is the technical term we use for the unified relational structure determined by Unifying Principle 2.12. All physically meaningful quantities are relational features of WILL.*

The Last Geocentric Epicycle

Definition 2.14 (Ontological weight)**.** *The number of irreducible primitives a theory posits.*

Picture	**Primitives**
Substantival: container + content (spacetime, energy/matter)	≥ 2
Relational: WILL (one self-relating structure)	1

It is the operation physics has performed at every unification:

- **Heat** $\equiv$ molecular motion (deletes caloric);
- **Electricity** $\equiv$ magnetism (Maxwell);
- **Space** $\equiv$ time (Einstein).

Each merged two primitives into one. **SPACETIME** $\equiv$ **ENERGY** applies the identical operation. **This Principle does not add, it subtracts: it removes the hidden assumption. Structure and dynamics are two aspects of a single entity that we call WILL.**

Remark 2.15 (Auditability)**.** *Principle 2.12 is foundational but testable: it is subject to: (i) geometric audit (internal logical consequences) and (ii) empirical audit (agreement with empirical data).*

3 Stage II: Geometric Derivation

3.1 WILL Structure

Having established Principle 2.12 by removing the illicit separation of structure and dynamics, we now proceed to derive its necessary geometric and physical consequences.

The foundational core (2.1.1)—Closure, Conservation, and Isotropy—establishes the necessary properties for the geometric carriers of the relational energy resource.

Relational Carrier Conventions:

All references to "carriers" within WILL RG open research are to be read in the strict relational sense:

- **Degree of freedom (DOF):** An n-dimensional relational carrier encoding the conserved transformation resource.
- **Direction:** An oriented relation. Opposite orientations are physically distinct and not identified.
- **Closed carrier:** Compact and without boundary; resource cannot leak into an external reservoir.
- **No background:** No external embedding space. All geometric structure is reconstructible solely from relations between participants.
- **Maximal symmetry:** The carrier is homogeneous and isotropic with respect to oriented directions.
- **Minimal relational carrier:** A closed, maximally symmetric carrier utilizing exactly the required DOF without arbitrary parameters.

3.2 Deriving Relational Carriers

Theorem 3.1 (Minimal Relational Carriers of the Conserved Energy Resource)**.** *The minimal relational carriers satisfying Closure, Conservation, Maximal Symmetry, and Mathematical Transparency are:*

(a) S^1 for directional (Kinematic) relational transformation;

(b) S^2 for omnidirectional (Potential) relational transformation.

Proof. The proof classifies the minimal types of relations and applies the derived geometric constraints:

(a) **1-DOF (Kinematic) Relation:** The minimal non-trivial relation involves 1 degree of freedom (1-DOF): a sequence of state transformations.

By Theorem 2.5 (Closure), this sequence must form a loop to prevent resource leakage. A discrete sequence (a finite cyclic group C_n) requires assigning a specific integer value n to the number of states. Principle 2.4 (Mathematical Transparency) prohibits the introduction of arbitrary numerical parameters without unique physical justification. To achieve a parameter-free structure, the state cardinality must remain undefined, forcing the continuum limit.

By Theorem 2.9 (Isotropy), this continuous 1-DOF loop must be maximally symmetric. The unique connected, closed, parameter-free 1-carrier is the continuous S^1 (circle).

(b) **2-DOF (Potential) Relation:** The next minimal relation involves 2 degrees of freedom (2-DOF): the omnidirectional distribution of the transformational resource from a primary state.

This requires a 2-dimensional carrier. By Principle 2.4, any discrete tiling of this distribution introduces arbitrary parameters (such as the number of nodes or faces), strictly forcing a continuous 2-carrier.

By Theorem 2.5, this continuous 2-carrier must be closed. By Theorem 2.9, it must be maximally symmetric (an isotropic distribution of potential).

Topological classification of a closed 2-carrier under the strict requirement of maximal continuous symmetry eliminates all anisotropic carriers (e.g., the torus T^2, which possesses preferred intrinsic axes). The unique parameter-free, maximally symmetric 2-carrier is S^2 (the 2-sphere).

Ontological Minimalism and Mathematical Transparency uniquely enforce S^1 and S^2 as the minimal relational carriers without invoking metric geometry *a priori*. □

WARNING: The Spatial Container Fallacy

S^1 and S^2 are strictly not to be interpreted as physical geometries embedded in a spacetime container.
Do not visualize a sphere or a ring expanding through a 3D void.

- S^2 has no physical surface area, no volume, and does not "dilute" a flux across a spatial distance.
- They are purely abstract, algebraic phase spaces—relational carriers encoding the closure, conservation, and isotropy of the transformational resource.

Spatial distance (r) and time are purely emergent descriptive labels that observers attach to these underlying algebraic phase patterns. The phase projection dictates the space; the space does not dictate the phase.

Remark 3.2 (Constructive Derivation Protocol)**.** *This derivation proceeds by construction from traced origins. Each object entering the chain - the principles, the theorems, the carriers - has an explicit derivation path. Alternative structures (discrete groups, higher-dimensional carriers, non-orientable surfaces) are not excluded by enumeration; they are absent because no derivation chain within this framework produces them.*

Logical Chain Summary

Methodological Constraints
$\downarrow$ *(apply to existing physics)*
False Separation: $\underbrace{\text{fixed manifold} + \text{metric}}_{\text{structure}} + \underbrace{\text{fields} + \text{constants}}_{\text{dynamics}}$
$\downarrow$ *(by Relational Origin Principle no background allowed)*
SPACETIME ≡ ENERGY
$\downarrow$ *(derive consequences)*
Closure + Conservation + Isotropy
$\downarrow$ *(derive primal relational carriers)*
S^1 *(1-DOF kinematic)* + S^2 *(2-DOF potential)*

3.3 The Amplitude-Phase Duality

The manifestation of any system is distributed between its internal (Phase) and external (Amplitude) aspects. This single geometric constraint gives rise to the core phenomena of modern physics:

Lemma 3.3 (Duality of Relation)**.** *By Relational Origin Principle (2.2) the identification of spacetime with energy and its transformations necessitates two complementary relational measures:*

1. *the* ***amplitude*** *of transformation (external shift), and*
2. *the* ***phase*** *of transformation (internal order).*

Proof. By Principle of Relational Origin (2.2) any complete description of transformation must specify both what changes and how that change is internally ordered. A single measure cannot capture both. The Relational Carriers S^1 and S^2 provide the minimal geometry enforcing such complementarity: their orthogonal projections furnish precisely two non-redundant, coupled axis. □

The orthogonal decomposition of the relational carriers S^1 and S^2 reveals a functional duality. Physical state is a combination of two projections:

Definition 3.4 (The Amplitude Projection (External Interaction))**.** *Denoted by β (kinematic) and κ (potential). This component measures the extent of external relation. It manifests as momentum or potential intensity. Physically, it represents the system's relational "shift" from the* ***Observer's Relational Origin*** *(the rest frame).*

$$\textit{Amplitude} \rightarrow \textit{External Power (Kinetic/Potential)}$$

Definition 3.5 (The Phase Projection (Internal Evolution))**.** *Denoted by β_Y and κ_X. This component measures the internal ordering. It governs the intrinsic scale of proper time and proper length. A Phase of 1 represents maximal internal flow (rest), while a Phase of 0 represents the collapse of internal causality (light-speed/event horizon).*

$$\textit{Phase} \rightarrow \textit{Internal Order (Time/Structure)}$$

Theorem 3.6 (Conservation of Relation)**.** *For both kinematic (S^1) and potential (S^2) modes, the sum of the squared Amplitude and squared Phase is strictly invariant:*

$$\underbrace{\textit{Amplitude}^2}_{\textit{External Interaction}} + \underbrace{\textit{Phase}^2}_{\textit{Internal Existence}} = 1 \tag{1}$$

Proof. By Theorem 3.1, the conserved relational resource is carried by the maximally symmetric geometries S^1 and S^2. Normalizing the total conserved resource to unity (1), any physical state corresponds to a point on these carriers. The algebraic identity of orthogonal projections on a unit circle and sphere dictates that the sum of their squares must equal the squared radius. Therefore, the parameters satisfy $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$, encoding the finite, closed relational budget. □

The Projections are Cross-Cultural Invariants

- S^1 is the minimal geometric structure that can encode a directional relation without absolute space. Its parameter is an angle θ_1 whose cosine ($\beta = \cos(\theta_1)$) we identify with the fraction v/c and whose sine ($\beta_Y = \sin(\theta_1)$) governs the internal ordering (proper time fraction).
- S^2 is the minimal geometric structure that can encode omnidirectional relation without absolute space. Its parameter is an angle θ_2 whose sine ($\kappa = \sin(\theta_2)$) we identify with the escape velocity fraction v_e/c and whose cosine ($\kappa_X = \cos(\theta_2)$) governs the internal ordering (proper time fraction).

Any two observers assign β and κ the same value for the same state, whatever their unit system — or even their descriptive vocabulary.

$$\beta = \frac{v}{c}, \qquad \kappa = \frac{v_e}{c} = \sqrt{\frac{R_s}{r}} = \sqrt{\frac{\rho}{\rho_{max}}}$$

are *translations* into descriptive vocabulary.

3.3.1 Consequence: Relativistic Effects

Proposition 3.7 (Physical Interpretation: Relativistic Effects)**.** *The conservation law of Theorem 3.6 implies that any redistribution between the orthogonal components manifests physically as the relativistic effects of time dilation and length contraction.*

Proof. By Theorem 3.6, the components satisfy $\beta^2+\beta_Y^2 = 1$ and $\kappa^2+\kappa_X^2 = 1$. An increase in Amplitude (β, κ) enforces a geometric decrease in Phase measure (β_Y, κ_X). This necessary reduction of β_Y and κ_X corresponds precisely to the dilation of proper time and the contraction of proper length, while the growth of β and κ represents momentum or potential depth. Thus, the kinematic and potential trade-off is the direct, un-parameterized physical expression of the geometric closure of the S^1 and S^2 carriers. □

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This identifies Relativistic and Gravitational Time Dilation not as a mysterious "slowing down" of clocks, but as geometric **phase rotation**. As a system invests more of its relational existence into external Amplitude (β or κ), it necessarily withdraws from its internal Phase (β_Y or κ_X), changing the rate of its proper time evolution.

Summary:

The geometry of spacetime is dictated by the Relations of energy states.

3.4 Relational Origin of the Inverse-Square Law and the Nature of Spatial Distance

Spatial distance is derived as the operational inverse of relational phase capacity. Treating distance as an *a priori* foundational primitive violates the principles of Relational Origin (2.2) and Ontological Minimalism (2.3).

Theorem 3.8 (Inverse-Square)**.** *The inverse-square potential proportionality $\kappa^2 \propto 1/r$ is a geometric necessity of the S^2 carrier's relational capacity, determined without spatial primitives.*

Proof. By the Spectroscopic Phase Shift Theorem, the potential projection κ^2 is operationally extracted entirely from the dimensionless phase tension (z_κ), independent of spatial parameters. Distance must therefore be defined exclusively by the relational tension between energy states.

Normalizing the absolute geometric scale at saturation ($\kappa^2 = 1 \equiv r = R_s$), spatial distance r emerges as the reciprocal measure of the omnidirectional projection amplitude on S^2:

$$\kappa = \sin(\theta_2) \implies \kappa^2 = \frac{R_s}{r} \implies r = \frac{R_s}{\kappa^2} \implies \kappa^2 \propto 1/r. \tag{2}$$

This is a direct realization of the Unifying Principle (2.12). The $1/r$ proportionality is an analytic identity defining spatial distance, not an empirical input. □

Summary

The inverse-square proportionality does not exist because physical space dictates how forces dilute. It exists because "spatial distance" is the descriptive vocabulary assigned to the inverse capacity of the omnidirectional relational carrier S^2.

3.5 Kinetic Energy Projection β on S^1

- **The Amplitude Component ($\beta = v/c$):** External shift or the extent of kinematic relation observed from an arbitrary relational origin (the observer's rest frame). It quantifies how much of the universal "transformation rate" is perceived as motion through space, relative to the observer. In legacy units, this is identified as the dimensionless ratio of velocity to the speed of light.
- **The Phase Component ($\beta_Y = \sqrt{1-\beta^2}$):** Internal ordering or the intrinsic scale of proper time and proper length within the moving system. A phase of 1 signifies maximal internal flow (rest), while a phase of 0 indicates the collapse of internal causality (light-speed).

These two components are bound by the conservation theorem (3.6) in closed system, which acts as a finite "budget of transformation":

$$\beta^2 + \beta_Y^2 = 1$$

The manifestation of any system is distributed between its internal (Phase) and relational (Amplitude) aspects.

Since S^1 encodes one-dimensional shift, the total energy E of the system must project consistently onto both external and internal axes:

$$E^2 = E_X^2 + E_Y^2$$

where

$$E_X = E\beta, \qquad E_Y = E\beta_Y.$$

Desmos project

Kinetic projection β on the S^1 carrier

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3.5.1 Geometric Nature of Rest Energy State

Theorem 3.9 (Invariant Projection of Rest Energy)**.** *For any state (β, β_Y) on the relational circle S^1, the total energy E must scale such that its internal projection remains constant and equal to the rest energy state E_0.*

$$E\beta_Y = E_0.$$

Proof. By the Relational Origin Principle (2.2) and Conservation Theorem (3.6), a system's internal energy measured in its own rest frame ($\beta = 0 \implies \beta_Y = 1$) is its rest energy, E_0. For another frame B observing system A, if A is in relative motion ($\beta > 0$), the total energy E differs, but the internal energy E_Y must remain an invariant property of A. To preserve this invariant and agree on measurements, frame B must measure A's internal phase projection as identical to the rest energy:

$$E_{Y_{B\leftarrow A}} = E_{Y_{A\leftarrow A}} = E_0. \tag{3}$$

Therefore, the internal energy E_Y measured externally by any frame remains constant and equivalent to the rest energy:

$$E_Y \equiv E_0, \tag{4}$$

where E_0 is the energy measured in the frame where the system is at rest ($\beta = 0$, $\beta_Y = 1$). Geometric consistency therefore requires the vertical leg to be fixed:

$$E_Y = E\beta_Y = E_0 \implies E = \frac{E_0}{\beta_Y}. \tag{5}$$

□

Summary:

The historical Lorentz factor γ is the reciprocal of β_Y. $\gamma = 1/\beta_Y$

3.5.2 The Geometric and Historical Nature of Mass

Corollary 3.10 (Rest Energy and Mass Equivalence)**.** *This geometry does not require mass as an independent concept. The invariant rest energy E_0 is numerically identical to mass m in natural units ($c = 1$), and is simply expressed in anthropocentric units (kilograms) via the conventional identity:*

$$E_0 = mc^2. \tag{6}$$

Proof. From the invariant projection $E\beta_Y = E_0$ and closure of S^1, no additional scaling parameter is required. Hence the conventional bookkeeping identities $E_0 = mc^2$ or $m = E_0/c^2$ reduce to tautologies. Mass is therefore not independent, but the rest-energy invariant itself. □

Summary:

Mass is the invariant projection of total rest energy expressed in anthropocentric units of kilograms.

3.5.3 Energy–Momentum Relation

Proposition 3.11 (Horizontal Projection as Momentum)**.** *On the relational circle, the unique linear geometric projection measure of external amplitude from rest is the horizontal projection $E_X = E\beta$; hence*

$$p \equiv E_X \equiv E\beta \quad (c = 1).$$

Proof. The rest state is $(\beta, \beta_Y) = (0, 1)$. A shift measure must (i) vanish at rest, (ii) grow monotonically with $|\beta|$, and (iii) flip sign under $\beta \mapsto -\beta$. The only relational candidate satisfying (i)-(iii) is the horizontal projection $E_X = E\beta$. Thus the identification is necessary and unique within methodological constraints. □

Desmos project

Energy-momentum Triangle

Corollary 3.12 (Energy–Momentum Relation)**.** *With p identified by Proposition 3.11 and $m = E_0$, the closure identity yields*

$$E^2 = p^2 + m^2 \quad (c = 1).$$

Equivalently, upon restoring c,

$$E^2 = (pc)^2 + (mc^2)^2.$$

Proof. By Conservation Theorem (3.6), $(E\beta)^2 + (E\beta_Y)^2 = E^2$. Substituting $p = E\beta$ and $m = E_0$ proves the claim. Restoring c is dimensional bookkeeping: $p \mapsto pc$ and $m \mapsto mc^2$, while E remains E, yielding the standard form. □

Remark 3.13 (Geometric Forms)**.** *The same identity may be expressed explicitly in terms of circle coordinates:*

$$E^2 = (\tfrac{\beta}{\beta_Y} E_0)^2 + E_0^2 = \big(\cot(\theta_1)\, E_0\big)^2 + E_0^2.$$

These are equivalent renderings of the same geometric necessity.

Remark 3.14 (Units sanity check - bookkeeping)**.** *In natural units ($c = 1$), the identification is $p \equiv E\beta$. Restoring c via dimensional bookkeeping ($p \mapsto pc$) yields:*

$$pc = E\beta = \frac{E}{c}v \implies p = \frac{E}{c^2}v. \tag{7}$$

With $E = \gamma mc^2$, this reduces to $p = \gamma mv$, the standard relativistic momentum. No new parameters are introduced.

$\beta = v/c \quad \theta_1 = \arccos(\beta)$	
Algebraic Form	**Trigonometric Form**
$\beta = v/c = \sqrt{1 - \beta_Y{}^2}$	$\beta = \cos(\theta_1)$
$\beta_Y = \sqrt{1 - (v/c)^2} = \sqrt{1 - \beta^2}$	$\beta_Y = \sin(\theta_1)$

Table 1: Geometric representation of relativistic effects.

Summary

The energy–momentum relation $E^2 = (pc)^2 + (mc^2)^2$ is a geometric identity of S^1 carrier.

3.6 Minkowski Interval as Inflation of the S^1 Closure

In legacy Special Relativity the Minkowski interval is posited as the physical metric of a 4D background manifold. We reverse the order of explanation. The primitive object is the dimensionless kinematic closure on the relational carrier S^1 (Theorem 3.6):

$$\beta^2 + \beta_Y^2 = 1. \tag{8}$$

This is a single, parameter-free relation between the external (amplitude) and internal (phase) projections of the conserved transformation budget. It invokes no coordinates, no container, and no external time. The Minkowski interval is *not* more fundamental than (8); it is what (8) becomes once a sequence of substantival posits is **added**, none of which is required by the relations themselves.

The inflationary posits. Beginning from the closure (8), legacy formalism introduces, in order:

(i) **A container.** The relation is declared to unfold *within* a pre-existing spatial manifold.

(ii) **Spatial coordinates.** Orthogonal labels x, y, z are imposed on the container to localize the amplitude projection.

(iii) **Externally flowing time.** An independent coordinate time t is posited and granted an autonomous flow, demoting the internal phase β_Y to a mere ratio of proper to coordinate time, $\beta_Y \equiv d\tau/dt$.

(iv) **A coordinate reference scale.** The combination c^2dt^2 is adopted as the magnitude against which the conserved budget is measured.

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The inflation. Under these posits the projections take their coordinate forms. The amplitude becomes the coordinate speed fraction distributed over the three container axes,

$$\beta^2 = \frac{v^2}{c^2} = \frac{dx^2 + dy^2 + dz^2}{c^2\, dt^2}, \tag{9}$$

and the phase becomes the time ratio $\beta_Y = d\tau/dt$. Substituting both into the closure (8),

$$\frac{dx^2 + dy^2 + dz^2}{c^2\, dt^2} + \left(\frac{d\tau}{dt}\right)^2 = 1, \tag{10}$$

multiplying both sides by c^2dt^2:

$$\frac{dx^2 + dy^2 + dz^2}{c^2dt^2} + \left(\frac{d\tau}{dt}\right)^2 = 1 \quad \xrightarrow{\times c^2dt^2} \quad dx^2 + dy^2 + dz^2 + c^2d\tau^2 = c^2dt^2$$

and clearing the posited denominator c^2dt^2 returns

$$c^2d\tau^2 = c^2dt^2 - dx^2 - dy^2 - dz^2. \tag{11}$$

This is the Minkowski interval exactly. The legacy translation mapping is therefore:

- **Kinematic amplitude:** $\beta \equiv \cos\theta_1 = \dfrac{v}{c}$;
- **Internal phase:** $\beta_Y \equiv \sin\theta_1 = \dfrac{d\tau}{dt} = \dfrac{1}{\gamma}$.

Epistemological Consequence: the Interval is the Inflated Form

The Minkowski interval is recovered without remainder — but only after four substantival structures (a container, three spatial coordinates, an autonomously flowing time, and a coordinate reference scale) have been appended to $\beta^2 + \beta_Y^2 = 1$. Not one of these is demanded by the relations; each is surplus ontology. The Lorentz factor $\gamma = 1/\beta_Y = 1/\sin\theta_1$ and kinematic time dilation ($\beta_Y < 1$) are accordingly not deformations of a spatial fabric but a phase rotation on the unit circle S^1. The interval carries more structure than the relation it encodes; the relation, not the interval, is the irreducible fundamental primitive.

3.7 Potential Energy Projection κ on S^2

IMPORTANT:

Throughout this work, S^1 and S^2 are not to be interpreted as spacetime geometries but purely as relational carriers encoding energy conservation. Any reading otherwise is a misinterpretation.

Analogous to S^1, the relational geometry of the sphere S^2 provides orthogonal projections for two aspects of omnidirectional transformation. We define them as follows:

- **The Amplitude Component ($\kappa = \frac{v_e}{c} = \sqrt{\frac{R_s}{r}}$):** The *relational potential measure* between the object and the observer. It corresponds to the *extent* of transformation as potential depth. Same way as we did with β on S^1, we will map κ on S^2 as fraction of speed of light. A value of $\kappa = 1$ denotes *saturation*: the entire relational resource of the system has been allocated into the potential channel. This condition mathematically defines the relational horizon where the escape velocity is equal to speed of light ($v_e = c$) or equivalently as radial distance equal to Schwarzschild radius ($r = R_s$ event horizon).
- **The Phase Component (κ_X):** This projection governs the intrinsic scale of proper length and proper time units within the potential field, corresponding to the internal *sequence* of transformation.

These two components are bound by the conservation theorem (3.6) in closed system, which acts as a finite "budget of transformation":

$$\kappa_X^2 + \kappa^2 = 1$$

The manifestation of any system is distributed between its internal (Phase) and relational (Amplitude) aspects. This single geometric constraint gives rise to the core gravitational phenomena of modern physics.

Desmos project

Potential projection as κ on S^2 carrier

3.7.1 Potential Meridional Section of S^2

By isotropy, the omnidirectional carrier is S^2, but any radially symmetric exchange reduces to a great-circle meridional section. We therefore work on a unit great circle of S^2 with the parameterization $(\kappa_X, \kappa) = (\cos\theta_2, \sin\theta_2)$.

3.7.2 Consequence: Gravitational Effects

The redistribution of the relational budget between the Phase and Amplitude components directly produces the effects of General Relativity. An increase in the relational amplitude measure (κ, gravitational potential) strictly necessitates a geometric decrease in the measure of the internal structure (κ_X). This geometric trade-off is observed physically as gravitational length contraction and time dilation. Thus, the curved geometry of spacetime is merely the shadow cast by the geometry of conserved relations.

3.7.3 Gravitational Tangent Formulation

Just as the relativistic energy–momentum relation can be expressed in terms of the kinematic projection $\beta = v/c$, we may construct its gravitational analogue using the potential projection $\kappa = v_e/c$, where v_e is the Newtonian escape velocity at radius r.

In the kinematic case, with $\beta = \cos\theta_1$, the energy relation can be written as

$$E^2 = \big(\cot\theta_1\, E_0\big)^2 + E_0^2, \tag{12}$$

so that the relativistic momentum is expressed as

$$p = (E_0/c)\cot\theta_1. \tag{13}$$

In full geometric symmetry, the gravitational case follows from $\kappa = \sin\theta_2$. We define the scalable gravitational total energy as

$$E_g = \frac{E_0}{\kappa_X}, \qquad \kappa_X = \sqrt{1-\kappa^2}, \tag{14}$$

and introduce the gravitational analogue of momentum (potential momentum):

$$p_g \;=\; (E_0/c)\tan\theta_2. \tag{15}$$

This yields the exact gravitational energy–momentum relation:

$$E_g^2 = (p_g c)^2 + (mc^2)^2. \tag{16}$$

Summary:

$$\beta = \cos\theta_1, \qquad \kappa = \sin\theta_2,$$

$$\beta \;\longleftrightarrow\; \kappa, \qquad \cot\theta_1 \;\longleftrightarrow\; \tan\theta_2.$$

Kinematic momentum p and gravitational momentum p_g are thus dual projections of the closed relational geometry, expressed through complementary trigonometric forms.

3.8 Schwarzschild Interval as Inflation of the S^2 Closure

The gravitational construction is strictly parallel to kinetic. The primitive object is the dimensionless potential closure on the relational carrier S^2 (Theorem 3.6):

$$\kappa^2 + \kappa_X^2 = 1, \tag{17}$$

relating the external potential amplitude κ to the internal phase κ_X. As before, (17) contains no radial coordinate, no central body, and no external time. The Schwarzschild component g_{tt} is recovered only by appending substantival posits.

The inflationary posits.

(i) **A container.** A spatial manifold is posited around a central body.

(ii) **A static observer.** The observer is held fixed in this container ($dr = d\theta = d\phi = 0$), so that only the temporal relation survives.

(iii) **Externally flowing time.** An independent coordinate time t is posited with autonomous flow, demoting the internal phase to the ratio $\kappa_X \equiv d\tau/dt$.

(iv) **A radial coordinate and legacy constants.** The dimensionless amplitude is localized as a coordinate function, $\kappa^2 \equiv R_s/r$, thereby introducing a radial coordinate r together with the legacy constants G and M through $R_s = 2GM/c^2$ — none of which is intrinsic to κ.

Note on Epistemic Hygiene. Posit (iv) is itself a legacy construction: κ is the primitive, while r, G, and M are coordinate-and-unit artifacts attached to it. In WILL R.O.M. (sections on relational parameterization, operational scale, and G) gravitational systems are described fully *without* G and *without* M, confirming that the projection κ, not R_s/r, is the irreducible primitive.

The inflation. Substituting $\kappa_X = d\tau/dt$ and $\kappa^2 = R_s/r$ into the closure (17),

$$\left(\frac{d\tau}{dt}\right)^2 + \frac{R_s}{r} = 1 \quad \Longrightarrow \quad \left(\frac{d\tau}{dt}\right)^2 = 1 - \frac{R_s}{r}, \tag{18}$$

and introducing the posited reference scale $c^2 dt^2$ gives

$$\kappa^2 + \kappa_X^2 = 1 \quad \Longrightarrow \quad c^2 d\tau^2 = c^2\left(1 - \frac{R_s}{r}\right) dt^2, \tag{19}$$

which is the Schwarzschild interval for a static observer ($ds^2 = c^2 d\tau^2$). The legacy translation mapping is:

- **Potential amplitude:** $\kappa \equiv \sin\theta_2 = \sqrt{\dfrac{R_s}{r}}$;
- **Internal phase:** $\kappa_X \equiv \cos\theta_2 = \sqrt{1-\kappa^2} = \dfrac{d\tau}{dt}$.

Remark 3.15 (Full derivation)**.** *This covers the static interval fixes the temporal projection $\kappa_X = d\tau/dt$. Full derivation of Schwarzschild and Kerr metrics we present within addition: R.O.M. (sec:Schwarzschild) , R.O.M. (sec:Kerr)*
.

Key Insight: Two Inflations

The Minkowski and Schwarzschild intervals are produced by the identical procedure: append a container, coordinates, an autonomously flowing time, and a coordinate reference scale to a single dimensionless closure — $\beta^2 + \beta_Y^2 = 1$ on the kinematic carrier, $\kappa^2 + \kappa_X^2 = 1$ on the potential carrier. Gravitational time dilation ($\kappa_X < 1$) is, like its kinematic counterpart, a phase rotation, not the curvature of a pre-existing fabric. What General Relativity treats as the fundamental metric of spacetime is the coordinate-laden image of a parameter-free relation. The relation is primary; the metric is its inflation.

3.9 Clear Relational Symmetry Between Kinematic and Potential Projections

On the unit kinematic circle (S^1) we parameterize

$$(\beta, \beta_Y) = (\cos\theta_1,\ \sin\theta_1),$$

so that the invariant internal phase projection reads

$$E\,\beta_Y = E_0 \Longrightarrow \boxed{E = \frac{E_0}{\beta_Y} = \frac{E_0}{\sin\theta_1}}, \qquad p = \frac{E}{c}\,\beta = \frac{E_0\,\beta}{\beta_Y} = E_0\cot\theta_1,$$

and therefore $E^2 = (pc)^2 + E_0^2$.

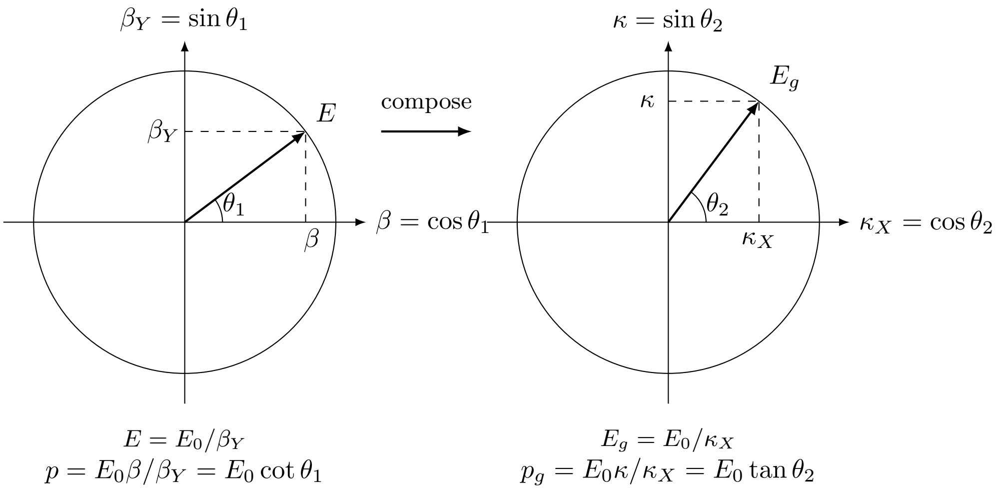
On the gravitational circle (S^2) we parameterize

$$(\kappa_X, \kappa) = (\cos\theta_2,\ \sin\theta_2),$$

so that the invariant internal phase projection reads

$$E_g\,\kappa_X = E_0 \Longrightarrow \boxed{E_g = \frac{E_0}{\kappa_X} = \frac{E_0}{\cos\theta_2}}, \qquad p_g = E_g\,\kappa = \frac{E_0\,\kappa}{\kappa_X} = E_0\tan\theta_2,$$

and therefore $E_g^2 = p_g^2 + E_0^2$.



Now we can clearly see the underlying symmetry between relativistic and gravitational factors that can be expressed in unified algebraic and trigonometric forms, as shown in Table 2.

3.10 Closure: Relational Origin of the Virial Theorem

Having established that directional (kinematic) and omnidirectional (potential) relations are carried by the S^1 and S^2 respectively, we now derive the relationship that unifies them.

3.10.1 Derivation of the Energetic Exchange Rate

Remark 3.16 (From slice to whole S^2)**.** *Although we parametrise a single meridional great circle* (κ_X, κ) *for algebraic convenience, the amplitude* κ^2 *denotes the* total omnidirectional budget *of the* S^2 *carrier. The exchange-rate factor* 2 *reflects that* S^2 *has two independent relational degrees of freedom even when calculations are carried out on a representative great-circle section.*

Remark 3.17 (Carrier capacities fix the rate)**.** *The exchange rate* 2 *is the ratio of the carriers' relational capacities, not a fitted constant. The one-DOF carrier* S^1 *saturates at* $\beta^2 = 1$ *(light,* $\beta_Y = 0$*); the two-DOF carrier* S^2 *saturates at* $\kappa^2 = 2$ *(only the extremal-Kerr state* $r = \frac{1}{2}R_s$*,* $\beta = 1$*). Equal quadratic weight per DOF (Lemma 3.18) gives* $\kappa^2_{\max}/\beta^2_{\max} = 2$*, identical to the closure rate* $\kappa^2 = 2\beta^2$*.*

3.10.2 Uniqueness of the Exchange Rate (No Hidden Weighting)

Lemma 3.18 (DOF-Indifference)**.** *Under maximal symmetry (no privileged directions, Theorem 2.9) and ontological minimalism (no hidden structure, Principle 2.3), any admissible conserved quadratic budget must assign* equal weight per independent relational degree of freedom. *This is the unique apportionment that introduces no additional parameters or distinctions beyond the carriers' degrees of freedom themselves.*

Proof. The conserved quadratic budget is a single scalar resource with no intrinsic scale. The only information available to apportion it is the number of independent relational degrees of freedom of each carrier. Any fixed ratio other than direct proportionality to the DOF count would require an extra rule that distinguishes the carriers on grounds other than their DOF. Such a rule would constitute hidden structure, violating all 4 methodological principles (2.2, 2.3, 2.4, 2.1).. Therefore, each independent DOF must receive identical quadratic weight b. □

Theorem 3.19 (Closure)**.** *In a relationally closed system, the kinematic carrier* S^1 *(1 DOF) and the potential carrier* S^2 *(2 DOF) exchange the conserved quadratic budget in proportion to their degrees of freedom. Consequently, any relationally closed system must satisfy*

$$\boxed{\kappa^2 = 2\beta^2}$$

globally.

Proof. Let b be the conserved quadratic budget associated with one independent degree of freedom. By Lemma 3.18, every DOF receives exactly this amount b.

The kinematic carrier S^1 possesses one degree of freedom, so its total amplitude budget is

$$B_{S^1} = 1 \cdot b = b.$$

The potential carrier S^2 possesses two degrees of freedom. Even when calculations are performed on a single meridional great-circle section for algebraic convenience, κ^2 represents the *total* omnidirectional budget of the full S^2 carrier. Its amplitude budget is therefore

$$B_{S^2} = 2 \cdot b.$$

An exchange rate $\mathcal{R}$ is defined by the requirement that the omnidirectional budget is a fixed multiple of the directional budget:

$$B_{S^2} = \mathcal{R}\, B_{S^1}.$$

Substituting the expressions above immediately yields $\mathcal{R} = 2$. By the conservation of the quadratic budget on each carrier (Theorem 3.6),

$$\kappa^2 = \mathcal{R}\beta^2 = 2\beta^2.$$

Any exchange rate other than $\mathcal{R} = 2$ would require either unequal weighting of degrees of freedom or an additional free parameter, both of which are forbidden by the four methodological principles (Principles 2.2, 2.3, 2.4, and 2.1).

$$\underbrace{1\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\text{Amplitude}^2}_{\text{on } S^1}$$

□

Definition 3.20 (Closure Factor)**.**

$$\delta \equiv \frac{\kappa^2}{2\beta^2}$$

For circular orbits, $\delta \equiv 1$. For elliptic orbits locally $\delta_{cucle} \neq 1$. For bound elliptical orbits $\langle\delta\rangle_{cycle} = 1$ to remain relationally closed.

Corollary 3.21 (Relational Closure Condition)**.** *Closed systems (momentary or periodic) satisfy $\kappa^2 = 2\beta^2$ identically. Open systems display $\langle\delta\rangle_{cycle} \neq 1$, the magnitude of which quantifies the energy flow through unaccounted channels. When all channels are included, closure is restored.*

Remark 3.22 (Physical Interpretation)**.** *The exchange rate between the kinematic and gravitational projections corresponds to the ratio of their relational dimensions. This purely geometric constant (2) replaces the empirical proportionalities of classical dynamics. It is the relational origin of the **virial theorem**: the kinetic and potential aspects of WILL maintain closure through the invariant DOF ratio.*

Illustrative Examples.

- **Circular Orbit (Closed).** A body at any orbital phase satisfies $\kappa^2 = 2\beta^2$. The entire conserved resource is partitioned between kinetic and gravitational projections; no internal "breathing" and no external channel exists.
- **Elliptical Orbit (Closed).** A body satisfies $< \kappa^2 >= 2 < \beta^2 >$ as an average per orbital cycle due to internal "breathing" of elliptical systems. Though this internal "breathing" is restricted by the Energy-Symmetry Law (3.12) so the difference $W = \frac{1}{2}(\kappa^2 - \beta^2)$ = constant at any orbital phase (R.O.M. sec:vis-viva). No external channel exists.
- **Radiating Binary (Open).** An elliptical compact binary violates $< \kappa^2 >= 2 < \beta^2 >$ when only orbital degrees of freedom are counted, the closure defect δ quantifying energy lost to gravitational radiation. Including all channels restores closure.

Summary:

1. WILL is a closed relational structure, SPACETIME ≡ ENERGY.
2. The simplest maximally symmetric carriers of these relations are S^1 and S^2.
3. The parameters $\beta = \cos\theta_1$ and $\kappa = \sin\theta_2$ are thus constrained to these carriers.
4. The geometric exchange rate between these modes equals the ratio of their DOF: 2.

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Remark 3.23 (Geometric Origin of Physical Law)**.** *The relation between kinetic and potential energy is not an empirical coincidence but a geometric necessity of relational closure. Classical mechanics merely approximates this deeper invariant. Explicitly,*

$$\boxed{\textit{Geometric Distribution } (\kappa^2) \;\equiv\; 2 \times \textit{Kinetic Distribution } (\beta^2).}$$

3.11 Total Relational Shift Q

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Q as Total Relational Shift

When an observer targets another system, the assigned Total Relational Shift norm Q is defined by the orthogonal projections:

$$\boxed{Q^2 = \beta^2 + \kappa^2} \tag{20}$$

Remark 3.24 (Closure-specific simplification)**.** *Under closure $\kappa^2 = 2\beta^2$ (circular/periodic systems), the norm reduces to $Q^2 = 3\beta^2$ (proved in section 3.10).*

In general (open or elliptic) configurations, the full definition with orbital phase "O" dependence $Q_o(O)^2 = \beta_o(O)^2 + \kappa_o(O)^2$ must be used. See R.O.M. paper for details and derivations.

3.11.1 Relational Reciprocity and Self-Centering

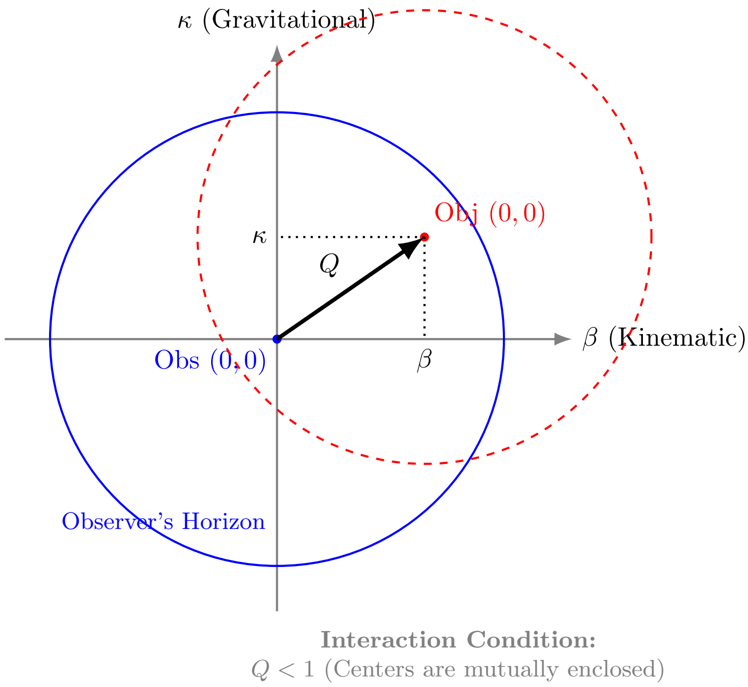

Figure 1: **Relational Self-Centering.** The total shift Q is defined by the orthogonal projections β and κ. Interaction is causal only when the center of the Object lies within the Observer's horizon ($Q < 1$), ensuring mutual coverage.

Theorem 3.25 (Relational Reciprocity)**.** *The operation of self-centering $(\beta, \kappa) = (0, 0)$ is a strict geometric necessity of relational origin principle (2.2), causal balance, and the Total Relational Shift Q between two systems is invariant regardless of the observing frame:*

$$Q^2_{A\to B} = Q^2_{B\to A}.$$

Proof. Consider two interacting systems, A and B. By relational origin principle (2.2) for observer A to measure the relational shift of B, system A must serve as the zero-state reference of its own local frame . This geometrically forces the self-centering operation:

$$(\beta_A, \kappa_A) = (0, 0).$$

From this relational origin, A measures the projections of B as (β_B, κ_B), yielding the norm:

$$Q^2_{A\to B} = \beta_B^2 + \kappa_B^2.$$

Conversely, when B observes A, system B constitutes the local zero-state $(\beta_B, \kappa_B) = (0, 0)$. This forces B to measure the exact same total shift norm for A:

$$Q^2_{B\to A} = \beta^2_A + \kappa^2_A = Q^2_{A\to B}.$$

Therefore, reciprocity is not a vector symmetry mapped onto a shared background space. It is the strict invariance of the relational norm Q under the self-centering operation required by relational origin principle (2.2). □

Summary

$$\boxed{\text{Relational reciprocity} \;=\; \text{invariance of the norm } Q \text{ under self–centering.}}$$

There is no common background arena. There are only mutual total shift magnitudes Q computed from each observer's own relational origin.

3.12 Energy-Symmetry Law

In RG, every transformation is bidirectional: each change observed by A corresponds to an equal and opposite change observed by B. This reciprocity is the algebraic form of causal continuity, and its geometric expression is the Energy-Symmetry Law.

3.12.1 Causal Continuity and Energy Symmetry

Theorem 3.26 (Energy Symmetry)**.** *The specific energy differences (per unit of rest energy) perceived by two observers for a transition between their states balance according to the Energy-Symmetry Law:*

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0. \tag{21}$$

Proof. Consider two observers:

- Observer A at rest on the surface at radius r_A (state defined by $\kappa_A, \beta_A = 0$).
- Observer B orbiting at radius $r_B > r_A$ with orbital velocity v_B (state defined by κ_B, β_B).

Each observer perceives energy transfers as the sum of the change in potential and kinetic energy budgets.

From A's perspective (transition from surface to orbit):

1. An object gains potential energy by moving away from the gravitational center.
2. It gains kinetic energy by accelerating to orbital velocity.

The total specific energy required for this transition is the sum of these two contributions:

$$\boxed{\Delta E_{A\to B} = \underbrace{\frac{1}{2}(\kappa^2_A - \kappa^2_B)}_{\text{Change in Potential}} + \underbrace{\frac{1}{2}(\beta^2_B - \beta^2_A)}_{\text{Change in Kinetic}}} \tag{22}$$

Since observer A is at rest, $\beta_A = 0$, and the expression simplifies to:

$$\Delta E_{A\to B} = \frac{1}{2}((\kappa^2_A - \kappa^2_B) + \beta^2_B) \tag{23}$$

From B's perspective (transition from orbit to surface):

1. The object loses potential energy descending into a stronger gravitational field.
2. It loses kinetic energy by reducing its velocity to rest.

This results in a specific energy difference:

$$\Delta E_{B\to A} = \frac{1}{2}((\kappa^2_B - \kappa^2_A) + (\beta^2_A - \beta^2_B)) = \frac{1}{2}((\kappa^2_B - \kappa^2_A) - \beta^2_B) \tag{24}$$

Summing these transfers gives:

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0 \tag{25}$$

Thus, no net energy is created or destroyed in a closed cycle of transitions, confirming the Energy-Symmetry Law as a direct consequence of the closed geometry. □

3.12.2 The Specific Energy Transfer (ΔE):

This is the projectional energy difference between states, equivalent to the Total Relational Shift Q, corresponding to the classical total energy of a transition (per unit rest energy). It is defined as the **sum of the changes** in the potential and kinetic energy budgets:

$$\Delta E_{A\to B} = \Delta U_{A\to B} + \Delta K_{A\to B} = \frac{1}{2}(\kappa_A^2 - \kappa_B^2) + \frac{1}{2}(\beta_B^2 - \beta_A^2) \tag{26}$$

It is this quantity, ΔE, that is conserved and must balance to zero in any closed system.

When the closure condition for stable, periodic orbits ($\kappa^2 - 2\beta^2 = 0$) is applied, the general Energy-Symmetry Law simplifies into remarkably elegant and direct forms. These simplified equations provide the precise energy balance for transitions involving energetically closed systems, such as planets or satellites in stable orbits.

Case 1: Surface-to-Orbit Transfer. For a transfer from a state of rest (A, where $\beta_A = 0$) to a closed orbit (B) where E_{0B} is the objects rest energy, the specific energy balance is given by:

$$\frac{E_{A\to B}}{E_{0B}} = \frac{1}{2}(\kappa_A^2 - \beta_B^2) \tag{27}$$

This result is derived by applying the closure condition $\kappa_B^2 = 2\beta_B^2$ to the general energy transfer formula, elegantly linking the initial potential projection to the final kinetic projection.

Case 2: Orbit-to-Orbit Transfer. For a transfer between two different closed orbits (A and B), the simplification is even more profound. The specific energy balance reduces to:

$$\frac{E_{A\to B}}{E_{0B}} = \frac{1}{2}(\beta_A^2 - \beta_B^2) \tag{28}$$

In this case, applying the closure condition to both the initial and final orbits causes the potential projection terms (κ^2) to cancel out completely. The entire energy balance of the transfer is expressed purely as the difference between the squares of the initial and final kinetic projections. This demonstrates a deep symmetry in the energetic structure of stable orbital systems.

3.12.3 Physical Meaning of the Factor $\frac{1}{2}$

The factor $\frac{1}{2}$ originate from the quadratic nature of the energy budgets in RG. The energetic significance of a state is proportional to the **square** of its geometric projection. This is the unavoidable consequence of relational carriers closure condition (amplitude^2 + phase^2 = 1). By using only amplitudes (β^2 and κ^2) we are operating with half the relational budgets of S^1 and S^2 carriers.

The individual energy budgets are:

- **Specific Potential Energy Budget:** $U/E_0 \propto -\frac{1}{2}\kappa^2$
- **Specific Kinetic Energy Budget:** $K/E_0 = \frac{1}{2}\beta^2$

The factor $\frac{1}{2}$ arises naturally when representing a conserved quantity (energy) through a quadratic measure (the square of a projection). The Energy-Symmetry Law deals with the sum of the *changes* in these individual budgets.

3.12.4 Universal Speed Limit as a Consequence of Energy Symmetry

Theorem 3.27 (Universal Speed Limit)**.** *The universal speed limit ($v \leq c$) emerges naturally from the requirement of energetic symmetry.*

Proof. Assume an object could exceed the speed of light, implying $\beta > 1$. In this scenario, its specific kinetic energy budget, $\frac{1}{2}\beta^2$, would become arbitrarily large.

The energy transfer required to reach this state, $\Delta E_{A\to B}$, would also become arbitrarily large. Consequently, no finite physical process could provide a balancing reverse transfer, $\Delta E_{B\to A}$, that would sum to zero. The fundamental symmetry would be broken:

$$\Delta E_{A\to B} + \Delta E_{B\to A} \neq 0 \tag{29}$$

Therefore, the condition $\beta \leq 1$ (which implies $v \leq c$) is an intrinsic requirement for maintaining the causal and energetic consistency of the relational universe. □

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3.12.5 Single-Axis Energy Transfer and the Nature of Light

Theorem 3.28 (Single-Axis Transformation). *For light, the kinematic projection amplitude reaches its full extent:*

$$\boxed{\beta = 1 \Longrightarrow \beta_Y = 0.}$$

This means that all transformation of the relational resource occurs along a ***single X axis*** $\Longrightarrow \beta = 1$*. The orthogonal Y axis is absent* $\Longrightarrow \beta_Y = 0$*, and the total resource of transformation is entirely expressed on one geometric component.*

Proof. For massive bodies, the Energy–Symmetry Law (Section 3.12) partitions the transformation resource equally between orthogonal axes. The specific binding energy invariant for orbital motion is therefore:

$$W_{\text{mass}} = \frac{1}{2}(\kappa^2 - \beta^2), \tag{30}$$

where the factor $\frac{1}{2}$ arises from this dual-axis distribution (Theorem 3.6). This invariant is conserved for closed systems and reduces to the classical Keplerian energy under the closure condition $\kappa^2 = 2\beta^2$ (Corollary 3.21).

Now consider light. By the Single–Axis Transformation Principle (Theorem 3.28), its kinematic projection saturates the carrier:

$$\beta = 1 \Longrightarrow \beta_Y = 0.$$

Geometrically, this collapses the relational architecture:

- The Y-axis (Phase component) vanishes entirely, as $\beta_Y = 0$ indicates no internal state evolution.
- No rest frame exists for self–centering (Section 3.11), eliminating the dual–framing that justifies the $\frac{1}{2}$ partitioning.
- The entire transformation resource concentrates on the single X-axis (Amplitude component).

Consequently, the energy invariant for photon interactions with a massive body (projection κ) must exclude the partitioning factor:

$$W_\gamma = \kappa^2 - \beta^2 = \kappa^2 - 1. \tag{31}$$

This is verified with no gravity state ($\kappa = 0$):

$$W_{\text{mass}} = \tfrac{1}{2}(0^2 - 0^2) = 0,$$
$$W_\gamma = 0^2 - 1^2 = -1.$$

The value $W_\gamma = -1$ corresponds to the full rest energy cost of transitioning to a photon state - whereas the $\frac{1}{2}$ factor would erroneously yield $-\frac{1}{2}$, violating energy symmetry.

For a concrete test case, consider a photon interacting with a massive body where $\kappa_M = 0.4$:

$$W_{\text{mass}} = \tfrac{1}{2}(0.4^2 - 1^2) = \tfrac{1}{2}(-0.84) = -0.42,$$
$$W_\gamma = 0.4^2 - 1^2 = -0.84 = 2 \times W_{\text{mass}}.$$

Thus, the gravitational effect on light is twice that on massive particles - matching the observed factor in light deflection.

The general energy potential follows directly: for light, the full geometric resource expresses unpartitioned along the single axis:

$$\Phi_\gamma = \kappa^2 c^2,$$

while massive bodies retain the partitioned form:

$$\Phi_{\text{mass}} = \tfrac{1}{2}\kappa^2 c^2.$$

This factor-of-2 is the geometric signature of axis count in relational space. No auxiliary approximations or background structures are introduced; the result emerges from the topological constraint $\beta_Y = 0$ applied to the closed carrier S^1. ∎

Summary

Energy Symmetry Law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$ - universal relational energy bookkeeping.
The Speed of Light is the boundary beyond which the energy symmetry law breaks down.
Causality is built-in feature of Relational Geometry.
Light has no rest frame. The disappearance of the Phase component (Y-axis $\beta_Y = 0$) concentrates the entire transformation resource on a single geometric component. This eliminates the $\frac{1}{2}$ partitioning factor, yielding $\boxed{\Phi_\gamma = \kappa^2 c^2}$ and explaining why light experiences twice the geometric effect of massive bodies.
This explains the experimentally verified factor of 2 in gravitational lensing and Shapiro delay, traditionally requiring full General Relativity to derive.

4 Stage III: Legacy Translation & Empirical Alignment

Logical Chain Summary

Methodological Constraints
$\downarrow$ *(apply to existing physics)*
False Separation: $\underbrace{\text{fixed manifold} + \text{metric}}_{\text{structure}} + \underbrace{\text{fields} + \text{constants}}_{\text{dynamics}}$
$\downarrow$ *(by Relational Origin Principle no background allowed)*
SPACETIME $\equiv$ ENERGY
$\downarrow$ *(derive consequences)*
Closure + Conservation + Isotropy
$\downarrow$ *(derive primal relational carriers)*
S^1 **(1-DOF kinematic) +** S^2 **(2-DOF potential)**
$\downarrow$ *(By Duality of Relation Lemma)*
Conservation of Relation $\underbrace{\text{Amplitude}^2}_{\text{External Interaction}} + \underbrace{\text{Phase}^2}_{\text{Internal Existence}} = 1$
$\downarrow$ *(By DOF-Indifference Lemma: equal quadratic weight to each independent DOF)*
$\kappa^2 = 2\beta^2 \equiv \underbrace{1\textbf{Amplitude}^2}_{\textbf{on } S^2} = \underbrace{2\textbf{Amplitude}^2}_{\textbf{on } S^1}$
$\downarrow$ *(By Relational Closure and Invariance theorems)*
Energy Symmetry Law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$

$\theta_1 = \arccos(\beta)$, $\theta_2 = \arcsin(\kappa)$, $\kappa^2 = 2\beta^2$, $a =$ semi-major axis	
Algebraic Form	**Trigonometric Form**
$\beta = v/c$	$\beta = \cos(\theta_1)$
$\kappa = \sqrt{R_s/a}$	$\kappa = \sin(\theta_2)$
$\beta_Y = \sqrt{1-\beta^2}$	$\beta_Y = \sin(\theta_1)$
$\kappa_X = \sqrt{1-\kappa^2}$	$\kappa_X = \cos(\theta_2)$
$p = E_0/c \cdot \beta/\beta_Y$	$p = E_0/c \cdot \cot(\theta_1)$
$p_g = E_0/c \cdot \kappa/\kappa_X$	$p_g = E_0/c \cdot \tan(\theta_2)$
$\tau = \beta_Y \kappa_X$	$\tau = \sin(\theta_1)\cos(\theta_2)$
$\tau_Y = \sqrt{1-\tau^2}$	$\tau_Y = \sqrt{1-\tau^2}$
$Q = \sqrt{\kappa^2+\beta^2} = \sqrt{3}\beta$	$Q = \sqrt{3}\cos(\theta_1)$
$Q_Y = \sqrt{1-Q^2} = \sqrt{1-\kappa^2-\beta^2} = \sqrt{1-3\beta^2}$	$Q_Y = \sqrt{1-3\cos^2(\theta_1)}$

Table 2: Unified representation of relativistic and gravitational effects for closed systems.

Summary

The familiar SR and GR factors emerge here as projections of the same conserved resource. Relativistic (β) and gravitational (κ) modes are not separate "effects" but dual aspects of one energy-transformation constraint revealing their unified origin.

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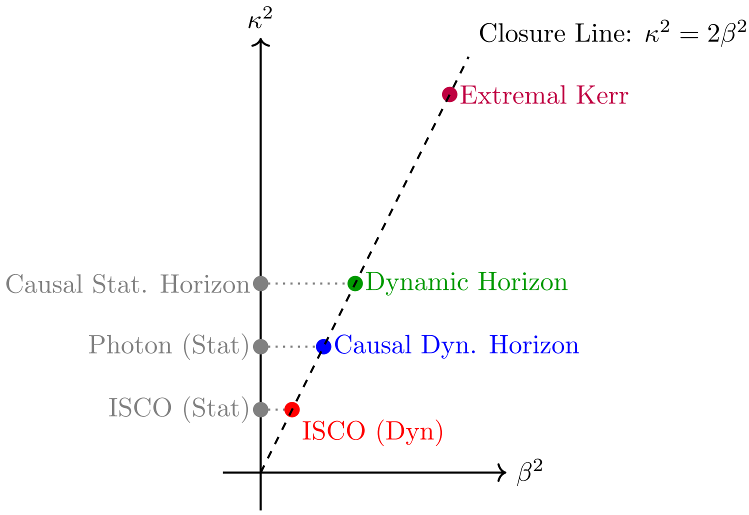
Phenomenon	Radius r	β^2	κ^2	Q^2	Comment
ISCO (Static location)	$3R_s$	0	$\frac{1}{3}$	$\frac{1}{3}$	Pure coordinate location, $r = R_s/\kappa^2$
ISCO (Dynamic state)	$3R_s$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	Marginal orbital stability, $Q^2 = Q_Y^2 = 1/2$
Photon sphere (Static)	$\frac{3}{2}R_s$	0	$\frac{2}{3}$	$\frac{2}{3}$	Pure coordinate location
Causal Dynamic horizon	$\frac{3}{2}R_s$	$\frac{1}{3}$	$\frac{2}{3}$	1	Relational horizon under closure $\kappa^2 = 2\beta^2$, $Q = 1$
Causal Static horizon	R_s	0	1	1	Purely gravitational closure limit, $\kappa^2 = 1$
Dynamic horizon	R_s	$\frac{1}{2}$	1	$\frac{3}{2}$	Orbital closure at Schwarzschild radius
Extremal Kerr horizon	$\frac{1}{2}R_s$	1	2	3	Maximal rotation, $\beta = 1$, merged horizons

Table 3: Critical radii and their projectional parameters. Static coordinate locations ($\beta = 0$) are strictly distinguished from their corresponding dynamic orbital states that emerge from the closure law $\kappa^2 = 2\beta^2$.



4.1 Equivalence Principle as Derived Identity

Lemma 4.1 (Unified Relational Scaling)**.** *Within the relational framework of WILL, both kinematic (S^1) and gravitational (S^2) transformations act as independent projections of the same invariant energy E_0. Each projection rescales the observable quantities by its respective geometric factor:*

$$E = \frac{E_0}{\beta_Y}, \qquad E_g = \frac{E_0}{\kappa_X}.$$

Proof. On the kinematic circle S^1, the invariant vertical projection corresponds to $\beta_Y = \sin\theta_1$. Preserving the same invariant leg E_0 forces the stretch $E/E_0 = 1/\beta_Y$. On the gravitational sphere S^2, the invariant horizontal projection is $\kappa_X = \cos\theta_2$, forcing $E_g/E_0 = 1/\kappa_X$. These transformations are independent and commute, each preserving the closure identity of its respective carrier. □

Theorem 4.2 (Equivalence of Inertial and Gravitational Response)**.** *Composing the independent relational stretches of Lemma 4.1 yields the total local energy scale*

$$E_{\text{loc}} = \frac{E_0}{\tau} = \frac{E_0}{\beta_Y\,\kappa_X} = \frac{E_0}{\sqrt{(1-\beta^2)(1-\kappa^2)}}.$$

The corresponding inertial and gravitational projections share a single operational factor,

$$\tilde{p} = \frac{E_{\text{loc}}}{c}\beta, \qquad \tilde{p}_g = \frac{E_{\text{loc}}}{c}\kappa,$$

both governed by the same effective mass

$$m_{\text{eff}} = \frac{E_0}{\beta_Y\,\kappa_X\,c^2} = \frac{E_0}{\tau c^2}.$$

Therefore,

$$\boxed{m_g \equiv m_i \equiv m_{\text{eff}}},$$

and the Einstein equivalence principle follows as a necessary structural identity of WILL.

Corollary 4.3 (Mass Invariance under Relational Scaling)**.** *The invariant core E_0 denotes the complete internal equilibrium state ($\beta_Y = \kappa_X = 1$). Relational factors β_Y and κ_X rescale only external manifestations (energy, momentum, and rates), while E_0 remains unchanged. Hence,*

$$\boxed{m_g \equiv m_i \equiv m = E_0/c^2},$$

is not a dynamical statement but the definition of rest invariance itself.

Remark 4.4 (Composition-Independence)**.** *Decomposing the invariant rest energy into internal channels,*

$$E_0 = \sum_a E_0^{(a)},$$

each term couples identically through the same geometric stretch:

$$E_{\text{loc}} = \sum_a \frac{E_0^{(a)}}{\tau}.$$

Since all channels scale by the same factor $1/\tau = 1/(\beta_Y \kappa_X)$, ratios between channels cancel in all observables. Therefore, composition-independence of motion (Eotvos universality) follows identically, without requiring a postulate $m_g = m_i$.

Remark 4.5 (Quantum Interface)**.** *The relational phase increment inherits the same scaling:*

$$\Delta\phi \propto E_{\text{loc}}\, \Delta\lambda,$$

where $\Delta\lambda$ is the internal ordering parameter. Thus both kinematic and gravitational phase shifts share the same stretch $1/\tau = 1/(\beta_Y \kappa_X)$, yielding composition-independent matter-wave interference patterns.

Summary:

In WILL, the equivalence of inertial and gravitational mass is not assumed but follows necessarily from the compositional closure of relational geometry. What General Relativity posits as a postulate, WILL reveals as a corollary.

4.2 Classical Mechanics

A striking consequence of the Energy–Symmetry Law (Section 3.12) emerges when analysing the total specific orbital energy. Since energy in RG is defined *relationally, as the measure of difference between two states*, we naturally select these two states (e.g., the surface of the central body 'A' and the orbit 'B') as the reference points for the potential and kinetic energy budgets. Under this relational approach, the total specific orbital energy (potential + kinetic, per unit rest mass) naturally appears in a form **structurally identical to the Minkowski interval**.

4.2.1 Classical Result with Surface Reference

For a test body of mass m on a circular orbit of radius a about a central mass $M_\oplus$ (Earth in our example), classical Newtonian mechanics gives:

$$\Delta U = -\frac{GM_\oplus m}{a} + \frac{GM_\oplus m}{R_\oplus}, \tag{32}$$

$$K = \frac{1}{2} m \frac{GM_\oplus}{a}. \tag{33}$$

Adding these and dividing by the rest–energy $E_0 = mc^2$ yields the dimensionless total:

$$\frac{E_{\text{tot}}}{E_0} = \frac{GM_\oplus}{R_\oplus c^2} - \frac{1}{2}\frac{GM_\oplus}{ac^2}. \tag{34}$$

4.2.2 Projection Parameters and Minkowski-like Form

Remark 4.6. *Here we explicitly define our projections using classical language to ease understanding for readers not yet familiar with WILL RG. For more explicit comparison and clearer derivations, we adopt standard Newtonian notation and define the projection parameters κ^2 and β^2 through their legacy equivalents involving G, M, and c. This is purely a pedagogical choice—the projections themselves remain dimensionless geometric quantities independent of these conventional constants.*

Define the WILL projection parameters for the surface and the orbit:

$$\kappa_\oplus^2 \equiv \frac{2GM_\oplus}{R_\oplus c^2}, \tag{35}$$

$$\beta_{\text{orbit}}^2 \equiv \frac{GM_\oplus}{ac^2}. \tag{36}$$

Substituting into (4.2.1) gives the exact identity:

$$\frac{E_{\text{tot}}}{E_0} = \frac{1}{2}\left(\kappa_\oplus^2 - \beta_{\text{orbit}}^2\right). \tag{37}$$

This is already in the form of a *hyperbolic difference of squares*: if we set $x \equiv \kappa_\oplus$ and $y \equiv \beta_{\text{orbit}}$, then

$$\frac{E_{\text{tot}}}{E_0} = \frac{1}{2}\,(x^2 - y^2), \tag{38}$$

which is structurally identical to a Minkowski interval in $(1+1)$ dimensions, up to the constant factor $\frac{1}{2}$.

Sign convention. We use $U/E_0 = -\frac{1}{2}\,\kappa^2$ and $K/E_0 = \frac{1}{2}\,\beta^2$ as *budgets*. The minus sign attaches to the potential budget by convention of reference (surface vs infinity); the budgets themselves are positive quadratic measures, while transfer ΔE is the signed sum of budget *changes*.

4.2.3 Physical Interpretation

In classical derivations, (4.2.1) is just the sum $\Delta U + K$ with a particular choice of potential zero. In the RG, (4.2.2) emerges directly from the energy–symmetry relation:

$$\Delta E_{A\to B} = \frac{1}{2}\big((\kappa_A^2 - \kappa_B^2) + \beta_B^2\big),$$

with $(A, B) = (\text{surface, orbit})$, and is *invariantly* expressible as a difference of squared projections.

This shows that the Keplerian total energy is not an isolated Newtonian artifact but a special case of a deeper geometric structure. While RG refuse to postulate any spacetime metric in the traditional sense, the emergence of this Minkowski-like structure from purely energetic principles is a powerful indicator of the deep identity between the geometry of spacetime and the geometry of energy transformation.

Why This Matters

- In classical form, the total orbital energy per unit mass depends only on GM and a, and is independent of the test–mass m.
- In WILL form, the same fact is embedded in Energy–Symmetry difference of squared projections, with no need for separate "gravitational" and "kinetic" constructs.
- This reframing answers *why* the Keplerian combination appears: it is enforced by the underlying geometry of energy transformation.

4.3 Two-Point Origin of the Lagrangian and Hamiltonian

In the standard formulation of mechanics, the Lagrangian $L = T - V$ and the Hamiltonian $H = T + V$ are treated as fundamental functions of a single configuration point $\mathbf{r}$. The Relational Geometry framework reveals that both are degenerate limits of a deeper two-point energy balance: the **Energy-Symmetry Law**

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0,$$

where the specific energy (per unit rest energy) perceived by an observer at state A for a transition to state B is

$$\boxed{\Delta E_{A\to B} = \tfrac{1}{2}\left(\kappa_A^2 - \kappa_B^2\right) + \tfrac{1}{2}\left(\beta_B^2 - \beta_A^2\right)}. \tag{39}$$

Here $\beta = v/c$ is the kinematic projection on the carrier S^1 and $\kappa^2 = R_s/r$ (with $R_s = 2GM/c^2$) is the potential projection on the carrier S^2. Both are directly measurable dimensionless ratios; the factor $\frac{1}{2}$ accounts for the fact that only half of each carrier's total relational budget is carried by the amplitude projection (the other half being the internal phase).

4.3.1 The clean relational Lagrangian

Choose a fixed reference state A – for instance, the surface of the central body – and consider a test body at a variable state B with velocity v and radial distance r. Since A is at rest on the surface, $\beta_A = 0$. The energy transfer from A to B is then

$$\Delta E_{A\to B} = \tfrac{1}{2}\big(\kappa_A^2 - \kappa^2(r)\big) + \tfrac{1}{2}\beta^2(v).$$

The quantity $\Delta E_{A\to B}$ is the energy that must be supplied to move the test body from the surface to the orbit. Interpreted as a Lagrangian, we may separate the kinetic and potential contributions and write the **relational Lagrangian.** The actual energy transfer is $mc^2\Delta E_{A\to B}$. Substituting the definitions immediately yields

$$mc^2\Delta E_{A\to B} = \tfrac{1}{2}mv^2 + \tfrac{1}{2}mc^2\kappa_A^2 - \frac{GMm}{r}$$

Interpreting the first two terms as the relational Lagrangian (with the constant surface term droppable) recovers the form used in mechanics.

$$L_{\rm rel} \;=\; \tfrac{1}{2}mv^2 \;+\; \tfrac{1}{2}mc^2\kappa_A^2. \tag{40}$$

The term $\frac{1}{2}mc^2\kappa_A^2$ is a constant (the surface potential budget) and does not affect the equations of motion; it can be dropped. Thus the clean relational Lagrangian is simply $L_{\rm rel} = \frac{1}{2}mv^2$, which is the Newtonian kinetic energy alone – the entire potential information is carried by the two-point difference, not by a pre-assigned function $V(r)$.

4.3.2 First ontological collapse: the Newtonian Lagrangian

The standard Newtonian Lagrangian is obtained by the **first ontological violation**: the two distinct points A and B are forced to coincide, $r_A = r_B = r$. This collapses the two-point structure into a single-point function. Formally, we set $\kappa_A^2 = \kappa^2(r)$ and $\beta_A = 0$, while keeping $\beta_B = \beta$. Equation (39) then becomes

$$\Delta E_{A\to B} = \tfrac{1}{2}\big(\kappa^2(r) - \kappa^2(r)\big) + \tfrac{1}{2}\beta^2 = \tfrac{1}{2}\beta^2,$$

which no longer contains any potential information. To recover a potential, one must *separately* introduce the concept of a potential energy function $V(r)$. The relational origin of $V(r)$ is lost; instead it is postulated as $V(r) = -GMm/r$. Adding this ad-hoc potential to the kinetic term yields the familiar Lagrangian

$$L \;=\; \tfrac{1}{2}mv^2 \;+\; \frac{GMm}{r},$$

which is exactly the expression obtained by taking the clean relational Lagrangian (40) and arbitrarily setting the constant surface term to GMm/r. Thus the Newtonian Lagrangian is the degenerate single-point limit of the two-point relational law.

4.3.3 Second ontological collapse: the Hamiltonian

If one starts from the same two-point law but reverses the sign convention for the potential budget (i.e., measures the potential from infinity downward), one obtains the total specific energy

$$E \;=\; -\tfrac{1}{2}\kappa^2 + \tfrac{1}{2}\beta^2.$$

Multiplying by mc^2 and substituting the definitions of κ and β gives

$$H \;=\; \tfrac{1}{2}mv^2 \;-\; \frac{GMm}{r},$$

which is the Newtonian Hamiltonian. The sign change reflects the switch from a surface-referenced budget (positive potential term) to an infinity-referenced budget (negative potential term). Both conventions are contained in the single two-point law; which one appears depends on which state is taken as the reference.

Two-point relational form	Collapsed single-point form
$\Delta E_{A\to B} = \frac{1}{2}(\kappa_A^2 - \kappa_B^2) + \frac{1}{2}(\beta_B^2 - \beta_A^2)$	$L = \frac{1}{2}mv^2 + \frac{GMm}{r}$ $H = \frac{1}{2}mv^2 - \frac{GMm}{r}$
The collapse is performed by identifying $r_A = r_B = r$ and then appending an ad-hoc potential function. The two-point law contains the full energy balance without the need for a separate potential.	

Table 4: Two-point relational law versus collapsed single-point form.

4.3.4 Summary

Key insight

The Lagrangian and Hamiltonian are not fundamental. They are artefacts of the single-point collapse of the two-point Energy-Symmetry Law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$. In the relational framework, all dynamical information is encoded in the dimensionless projections β and κ and their mutual closure $\kappa^2 = 2\beta^2$. No potential function, no force law, and no action principle are required.

Remark 4.7 (Mathematical Status: Groupoid vs. Group)**.** *From a category-theoretic perspective, the relational transitions $\Delta E_{A\to B}$ form a* ***Groupoid*** *structure, where operations are defined only between specific connected states. Standard field theories (Hamiltonian/Lagrangian) rely on the collapse of this structure into a* ***Group*** *acting on a global manifold (Quotienting). Thus, the relational Energy-Symmetry Law operates at the Groupoid level, prior to the reduction into global field theories.*

Legacy Dictionary (for conventional formalisms).
Within RG, all physical content is expressed purely in terms of relational projections β and κ on S^1 and S^2. For readers accustomed to standard frameworks, the following translation rules may help:

1. *General Relativity (metric form):*

$$\kappa_X \mathrel{\hat{=}} \sqrt{-g_{tt}} \quad \text{(static spacetimes)}, \qquad \beta \mathrel{\hat{=}} \frac{\|u^\mu_{\text{spatial}}\|}{u^t c}.$$

2. *Canonical mechanics (Lagrangian/Hamiltonian):* Quantities such as $p_i = \partial L/\partial \dot{q}^i$ do not belong to the ontology of RG. They arise only after collapsing the two-point relational law into a one-point formalism. They are secondary derived quantities arising after the two-point relational structure is reduced to a local single-point formalism.

Here the symbol $\hat{=}$ denotes not an ontological identity, but a pragmatic dictionary entry for translation into legacy notation.

4.3.5 Third Ontological Collapse: Newton's Third Law

We now demonstrate that Newton's Third Law, like the Lagrangian and Hamiltonian, emerges as a limiting case of RG. It arises as a necessary mathematical consequence of the same ontological collapse - forcing a two-point relational law into a single-point, instantaneous formalism.

Theorem 4.8 (Newton's Third Law as a Single-Point Limit)**.** *The Energy–Symmetry Law ($\Delta E_{A\to B} + \Delta E_{B\to A} = 0$) mathematically necessitates Newton's Third Law ($\vec{F}_{AB} = -\vec{F}_{BA}$) in the classical, non-relativistic limit where the two-point relational energy budget is collapsed into a single-point potential function $U(\vec{r})$.*

Proof. We begin with the Energy–Symmetry Law (Section 3.12), the principle of causal balance for state transitions:

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0.$$

In the classical, non-relativistic limit, this two-point relational law is "ontologically collapsed" into a single-point potential energy function, U. This function is assumed to depend only on the relative positions of the two interacting entities, A and B:

$$U = U(\vec{r}) \quad \text{where} \quad \vec{r} = \vec{r}_B - \vec{r}_A.$$

This $U(\vec{r})$ is the classical approximation of the system's relational energy budget. In this collapsed formalism, the force $\vec{F}$ is defined as the negative gradient of this potential.

(1) Force on B by A ($\vec{F}_{AB}$): This force is found by taking the gradient with respect to B's coordinates:

$$\vec{F}_{AB} = -\nabla_B U(\vec{r}_B - \vec{r}_A) \tag{41}$$
$$= -(\frac{dU}{d\vec{r}}) \cdot (\frac{\partial \vec{r}}{\partial \vec{r}_B}) \tag{42}$$
$$= -\nabla U(\vec{r}) \cdot (\mathbf{I}) \tag{43}$$
$$= -\nabla U(\vec{r}) \tag{44}$$

(2) Force on A by B ($\vec{F}_{BA}$): This force is found by taking the gradient with respect to A's coordinates:

$$\vec{F}_{BA} = -\nabla_A U(\vec{r}_B - \vec{r}_A) \tag{45}$$
$$= -(\frac{dU}{d\vec{r}}) \cdot (\frac{\partial \vec{r}}{\partial \vec{r}_A}) \tag{46}$$
$$= -\nabla U(\vec{r}) \cdot (-\mathbf{I}) \tag{47}$$
$$= +\nabla U(\vec{r}) \tag{48}$$

(3) Conclusion: By direct comparison of the results, we find:

$$\vec{F}_{AB} = -\nabla U(\vec{r}) \quad \text{and} \quad \vec{F}_{BA} = +\nabla U(\vec{r}).$$

Therefore, within the single-point collapsed formalism:

$$\boxed{\vec{F}_{AB} = -\vec{F}_{BA}}$$

This completes the proof. Since all physical observations are strictly relative, embedding the energy budget within an absolute spatial container ($\vec{r}_A, \vec{r}_B$) introduces unobservable parameters, which strictly violates the principle of Epistemic Hygiene. To remain empirically grounded, the collapsed potential U must depend exclusively on the measurable physical relation $\vec{r}_B - \vec{r}_A$. Consequently, the principle of equal and opposite forces emerges not as an empirical axiom of mechanics, but as the mathematical signature of any causally balanced, background-independent system $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$. □

4.3.6 Fourth Ontological Collapse: The Einstein-Hilbert Action (S_{EH})

The Einstein-Hilbert action ($S_{EH} = \frac{1}{16\pi G}\int R\sqrt{-g}\, d^4x$) represents a reduction of relational connections between states into localized coordinate points. Standard tensor calculus employs the limit $dx \to 0$ to compress relational tension into the local Ricci scalar (R). This operation necessitates the postulation of a smooth background manifold and the application of the Principle of Stationary Action ($\delta S_{EH} = 0$) to maintain global equilibrium across arbitrary coordinates.

4.3.7 Components Mapping

- **Global 4D Volume** ($\int \sqrt{-g}\, d^4x \to 1$): Instead of summation over a background manifold, relational capacity in RG is bound by the carriers $\kappa^2 + \kappa_X^2 = 1$ and $\beta^2 + \beta_Y^2 = 1$(3.6).
- **The Ricci Scalar** ($R \to \kappa^2, \beta^2$): Differential approximation replaced with relational projections.
- **The Variational Search** ($\delta S = 0 \to \Delta E_{A\to B} + \Delta E_{B\to A} = 0$): Reduced to the Law of Energy Symmetry ($\Delta E_{A\to B} + \Delta E_{B\to A} = 0$) (3.12).

Empirical Grounding

of the action (S_{EH}) manifests in gravitational light deflection via the addition of a curved spatial metric (g_{rr}). RG achieves the identical empirical result without introducing new ontological primitives: (Gravitational Deflection and Lensing).

Philosophical Consequences

Substantialism vs. Relationalism

4.4 Earth–GPS System: Standard GR as First-Order Approximation of RG

We made lots of derivations and daring claims. Its time to put them to the test. For our first tests we will use the standard Earth - GPS satellite system to challenge theorems and lemmas: (2.5; 2.7; 3.3; 3.6; 3.7; 3.9; ??; 3.11; 3.19; 3.12; 5.2; 4.3; 4.3.5)

4.4.1 Statement of the Verification Problem

The standard post-Newtonian expression for the secular time shift between a clock on the Earth's surface and a clock on a circular GPS orbit, accumulated over one solar day, is

$$\Delta t_{\mathrm{GR}} \;=\; \left(\frac{\Phi_{\mathrm{GPS}}-\Phi_E}{c^2} \;-\; \frac{v_{\mathrm{GPS}}^2-v_E^2}{2\,c^2}\right) D\,M, \tag{49}$$

with $D = 86{,}400\,\mathrm{s}$, $M = 10^6$, $\Phi(r) = -GM_\oplus/r$, $v_E = 2\pi R_E/D$, and $v_{\mathrm{GPS}} = \sqrt{GM_\oplus/r_{\mathrm{GPS}}}$.

WILL Relational Geometry (WILL RG) expresses the same time shift as an exact algebraic ratio of two-carrier projection invariants $\tau \equiv \sqrt{1-\kappa^2}\,\sqrt{1-\beta^2}$:

$$\Delta t_{\mathrm{RG}} \;=\; \left(1 \;-\; \frac{\sqrt{1-\kappa_E^2}\,\sqrt{1-\beta_E^2}}{\sqrt{1-\kappa_{\mathrm{GPS}}^2}\,\sqrt{1-\beta_{\mathrm{GPS}}^2}}\right) D\,M, \tag{50}$$

where $\kappa^2(r) = 2GM_\oplus/(rc^2)$, $\beta^2 = v^2/c^2$, and the orbital closure condition $\beta_{\mathrm{GPS}}^2 = \kappa_{\mathrm{GPS}}^2/2$ holds for circular motion.

The question addressed in this section is whether (49) is the leading-order Taylor expansion of (50) in the small parameters κ^2 and β^2. If so, the algebraic difference of the two coefficients must vanish identically at first order, and the residual at finite values of the small parameters must reproduce the analytically derived second-order correction.

4.4.2 Methodology

The verification is split into two independent parts, both executed in a single public notebook (Section 4.4.5).

Part I — symbolic identity. Introduce the dimensionless geometric ratio $\rho \equiv R_E/r_{\mathrm{GPS}}$, which is treated as a finite (order-unity) parameter and is *not* expanded. The exact WILL RG shift coefficient is then

$$\delta_{\mathrm{RG}}(\kappa_E^2, \beta_E^2, \rho) \;=\; 1 - \frac{\sqrt{(1-\kappa_E^2)(1-\beta_E^2)}}{\sqrt{(1-\kappa_E^2\rho)(1-\kappa_E^2\rho/2)}}. \tag{51}$$

A single bookkeeping parameter ε is introduced via $\kappa_E^2 \to \varepsilon\,\kappa_E^2$, $\beta_E^2 \to \varepsilon\,\beta_E^2$, and the Taylor series of (51) is computed to second order in ε using `sympy.series`. The first-order coefficient $\delta_{\mathrm{RG}}^{(1)}$ is then subtracted from the dimensionless form of the GR coefficient δ_{GR} obtained from (49) under the substitutions $\Phi/c^2 = -\kappa^2/2$ and $v^2/c^2 = \beta^2$. The result is simplified algebraically.

Part II — numerical comparison. Both (49) and (50) are evaluated with 40-digit precision arithmetic (`mpmath`, `mp.dps = 40`) using a single set of input parameters: $c = 299{,}792{,}458\,\mathrm{m\,s^{-1}}$, $GM_\oplus = 3.986004418 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$, $R_E = 6{,}378{,}137\,\mathrm{m}$, $r_{\mathrm{GPS}} = 26{,}561{,}750\,\mathrm{m}$, $D = 86{,}400\,\mathrm{s}$. The closure residual $\beta_{\mathrm{GPS}}^2 - \kappa_{\mathrm{GPS}}^2/2$ is monitored as a numerical consistency check. The numerical discrepancy $\Delta t_{\mathrm{RG}} - \Delta t_{\mathrm{GR}}$ is then compared with the analytically predicted second-order correction $\delta_{\mathrm{RG}}^{(2)}\,D\,M$ obtained from Part I.

4.4.3 Results

Symbolic. The first-order Taylor coefficient of the WILL RG shift, expressed in κ_E^2, β_E^2, and the geometric ratio ρ, is

$$\delta_{\mathrm{RG}}^{(1)} \;=\; \frac{\kappa_E^2}{2} \;+\; \frac{\beta_E^2}{2} \;-\; \frac{3\,\kappa_E^2\,\rho}{4}. \tag{52}$$

The GR coefficient, expressed in the same variables, is

$$\delta_{\mathrm{GR}} \;=\; \frac{\kappa_E^2}{2} \;+\; \frac{\beta_E^2}{2} \;-\; \frac{3\,\kappa_E^2\,\rho}{4}. \tag{53}$$

The algebraic difference $\delta_{\mathrm{RG}}^{(1)} - \delta_{\mathrm{GR}}$ vanishes identically.

The second-order content of (51), which is absent from the additive GR formula (49), is

$$\delta_{\mathrm{RG}}^{(2)} \;=\; \frac{\kappa_E^4}{8} \;+\; \frac{\beta_E^4}{8} \;-\; \frac{\beta_E^2\,\kappa_E^2}{4} \;+\; \frac{3\,\kappa_E^4\,\rho}{8} \;-\; \frac{19\,\kappa_E^4\,\rho^2}{32} \;+\; \frac{3\,\beta_E^2\,\kappa_E^2\,\rho}{8}. \tag{54}$$

Three structural features of (54) have no analogue in any rearrangement of the additive GR sum (49): (i) the pure quadratic contributions $\kappa_E^4/8$ and $\beta_E^4/8$ arising from higher-order expansion of $\sqrt{1-x}$; (ii) the cross-term $-\beta_E^2\kappa_E^2/4$, which originates from the multiplicative coupling of the gravitational and kinematic projection factors within τ_E/τ_{GPS}; and (iii) the ρ^2 contribution, which encodes nonlinear dependence on the relative depth of the gravitational well.

Numerical. With the parameter set listed above:

quantity	value [μs day^{-1}]
Δt_{RG} (exact)	38.5421472752
Δt_{GR} (first order)	38.5421472451
$\Delta t_{\text{RG}} - \Delta t_{\text{GR}}$	3.017×10^{-8}
$\delta_{\text{RG}}^{(2)} \cdot D\,M$ (predicted)	3.017×10^{-8}
residual	$\sim 10^{-17}$

The numerical discrepancy $\Delta t_{\text{RG}} - \Delta t_{\text{GR}}$ matches the analytically derived second-order correction $\delta_{\text{RG}}^{(2)}\,D\,M$ to within the limit set by the 40-digit floating-point arithmetic. The closure residual $\beta_{\text{GPS}}^2 - \kappa_{\text{GPS}}^2/2$ vanishes to $\sim 10^{-50}$. The relative magnitude of the term omitted by the GR additive formula is $\sim 7.8 \times 10^{-10}$ of the total shift.

4.4.4 Interpretation

Equations (52)–(53) establish that the additive post-Newtonian formula (49) coincides exactly with the linear Taylor coefficient of the WILL RG exact algebraic ratio (50). This is an algebraic identity in $(\kappa_E^2, \beta_E^2, \rho)$, not a numerical near-match: the additive sum-of-corrections structure of (49) is recovered as the unique leading-order content of the multiplicative ratio τ_E/τ_{GPS}.

For the Earth–GPS configuration, the second-order term (54) contributes at the level of $\sim 3 \times 10^{-8}\,\mu$s day^{-1}, i.e. $\sim 8 \times 10^{-10}$ of the total accumulated shift. This is below the present precision of operational GPS clock comparisons and does not, by itself, constitute a distinguishing observation between the two formulations in this regime. The structural difference between an additive and a multiplicative composition of projection factors — in particular the presence of the $\beta^2\kappa^2$ cross-term in (54) — becomes quantitatively relevant only when κ^2 or β^2 approach order unity.

The scope of the present statement is restricted to the relation between the WILL RG closed-form expression (50) and the *additive post-Newtonian* form of the GR time-dilation formula (49). A comparison against fully non-linear geodesic integration in the exact Schwarzschild geometry combined with special-relativistic kinematics for the rotating Earth surface is a separate computation and is not addressed in this section.

4.4.5 Reproducibility

The complete symbolic and numerical verification reported in this section is contained in a single public Google Colab notebook, comprising the two scripts used to produce (52), (54), and the numerical table above:

Desmos project

Earth-GPS time, energy symmetry and WILL invariant test

Colab Notebook

Earth-GPS 1st order RG=GR.ipynb

The notebook has no external dependencies beyond `sympy` and `mpmath` and is self-contained. All input parameters are listed explicitly in the numerical script; any change in r_{GPS}, R_E, or $GM_\oplus$ rescales both Δt_{RG} and Δt_{GR} consistently and does not affect the symbolic identity $\delta_{\text{RG}}^{(1)} \equiv \delta_{\text{GR}}$.

4.5 Relational Orbital Mechanics (R.O.M.)

Relational Orbital Mechanics (R.O.M.)

Natural evolution of this section within WILL RG Open Research lead it to branch in to its own but not separate paper Full R.O.M. paper. It still remain structurally and logically a crucial part of this document that caries most of empirical applications and computational tests.

You can access the paper using the link above, and for quick introduction here's the content:

Contents

Relational Orbital Mechanics as algebraically closed system of equations developed in this section demonstrates that the entire phenomenology of motion – from Keplerian architecture to perihelion precession and light deflection – emerges as an inevitable algebraic consequence of the Foundational Methodological Principles. All WILL RG results including the complete success of R.O.M. remains an unbroken chain of derivations from first principles under extreme methodological constraints with zero flexibility.

4.6 Density, Mass, and Pressure

4.6.1 Derivation of Density

Translating RG (2D) to Conventional Density (3D). In RG κ^2 is the 2D parameter defined in the relational carrier S^2 . In conventional physics, the source term is volumetric density ρ, a 3D concept defined by the "cultural artifact" (a Newtonian "cannonball" model) of mass-per-volume .

To compare with measurements that are historically expressed in 3D terms, we need to bridge our 2D theory with 3D units that empirical data is using. We must create a "translation interface". We do this by adopting the conventional (Newtonian) definition of density, $\rho \propto M/r^3$, as our "translation target".

From the projective analysis established in the previous sections:

$$\kappa^2 = \frac{R_s}{r},$$

where κ emerges from the energy projection on the area of unit sphere S^2, and $R_s = 2GM/c^2$ links to the mass scale factor $M = E_0/c^2$.

This leads to mass definition:

$$M = \frac{\kappa^2 c^2 r}{2G}$$

To translate this into a volumetric density, we first adopt the conventional 3D (volumetric) proxy, r^3. This is not a postulate of RG, but the first step in applying the legacy (3D) definition of density:

$$\frac{M}{r^3} = \frac{\kappa^2 c^2}{2Gr^2}$$

This expression, however, is incomplete. Our κ^2 "lives" on the 2D surface S^2 (which corresponds to 4π), while the r^3 proxy assumes a 3D volume (which corresponds to $4\pi/3$). To correctly normalise the 2D parameter κ^2 against the 3D volume, we must apply the geometric normalization factor of the S^2 carrier by dividing by the area of the sphere, which is $1/4\pi$ (we explicitly reject the Newtonian volume integration $(4\pi/3)$ in favor of the S^2 carrier's geometric surface (4π). This enforces the relational principle that structural capacity scales exclusively via the 2D spherical boundary of interaction, not by filling an assumed 3D volumetric container).

This normalization is the necessary geometric step to interface the 2D relational carrier (S^2) with the 3D legacy definition of density:

$$\rho = \frac{1}{4\pi}\left(\frac{\kappa^2 c^2}{2Gr^2}\right)$$

$$\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$$

Local Density $\propto$ Relational Potential Projection

Maximal Density. At $\kappa^2 = 1$ (the horizon condition (for non rotating systems), $r = R_s$), this density reaches its natural bound, $\rho_{\max}$:

$$\rho_{\max} = \frac{c^2}{8\pi G r^2}$$

Normalized Relation. Thus, our "translation" reveals an identity: the potential projection κ^2 is the ratio of density to the maximal density:

$$\boxed{\kappa^2 = \frac{\rho}{\rho_{\max}} \quad \rightarrow \quad \kappa^2 \equiv \Omega}$$

Self-Consistency Requirement

The mass scale factor can be expressed in two equivalent ways.

From the geometric definition:

$$M = \frac{\kappa^2 c^2 r}{2G}.$$

From the energy density:

$$M = \alpha\, r^n\, \rho.$$

Substituting $\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$ into $M = \alpha\, r^n\, \rho$ gives

$$M = \frac{\alpha \kappa^2 c^2 r^{n-2}}{8\pi G}.$$

Equating the two forms:

$$\frac{\alpha r^{n-2}}{8\pi} = \frac{r}{2}.$$

For the mass M to remain a constant independent of the measurement scale r, the exponent must be $n = 3$, yielding $\alpha = 4\pi$. Hence,

$$M = 4\pi r^3 \rho,$$

which closes the consistency loop between the geometric and density-based formulations.

Desmos Project

Derivation of Density and Pressure

4.6.2 Pressure as Surface Curvature Gradient

In the RG framework pressure is not a thermodynamic assumption but the direct consequence of curvature gradients. The radial balance relation gives

$$P(r) \;=\; \frac{c^4}{8\pi G}\,\frac{1}{r}\,\frac{d\kappa^2}{dr}.$$

Using $\kappa^2 = R_s/r$, one finds $d\kappa^2/dr = -\kappa^2/r$, hence

$$P(r) \;=\; -\,\frac{\kappa^2 c^4}{8\pi G r^2}.$$

Since the local energy density is

$$\rho(r) \;=\; \frac{\kappa^2 c^2}{8\pi G r^2},$$

this yields the invariant equation of state

$$\boxed{P(r) = -\,\rho(r)\,c^2}.$$

Interpretation. P is a surface-like negative pressure (isotropic tension), not a bulk volume pressure. It expresses the resistance of energy-geometry itself to changes in projection.

Consistency. If one formally freezes the projection parameter ($d\kappa^2/dr = 0$), then $P = 0$. But in this case the angular curvature terms remain uncompensated, and the field equation is no longer satisfied. Any nontrivial radial dependence of κ inevitably generates the negative tension

$$P = -\rho c^2,$$

which precisely cancels the residual curvature. Thus the negative pressure is not optional but a necessary ingredient for full self-consistency.

Desmos Project

Derivation of Density and Pressure

Maximum pressure. At the geometric bound $\kappa^2 = 1$ (horizon condition), the density saturates at

$$\rho_{\max} = \frac{c^2}{8\pi G r^2},$$

and the corresponding pressure is

$$P_{\max} = -\,\rho_{\max}\,c^2 \;=\; -\,\frac{c^4}{8\pi G r^2}.$$

This negative surface pressure represents the ultimate tension limit of spacetime fabric at a given scale r.

Pressure in WILL is the intrinsic surface tension of energy-geometry, saturating at $P_{\max} = -c^4/(8\pi G r^2)$.

Physical meaning

The negative pressure is not an exotic substance but the geometric tension required to maintain self-consistency when κ^2 varies radially. It's the relational analogue of surface tension in a soap bubble.

4.7 Unified Geometric Field Equation

From the energy-geometry equivalence, the complete description of gravitational phenomena reduces to a single algebraic relation linking the geometric scale to the energy density ratio:

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho_{field}}{\rho_{max}}}$$

This identity defines the **local energy state of the relational geometry itself**. Here $\rho_{\max} = c^2/(8\pi G r^2)$ is the saturation density limit, and ρ_{field} is the effective energy density of the relational curvature.

4.8 Field Equation and Matter Sources

For a static, spherically symmetric configuration containing matter with density $\rho_{\text{matter}}(r)$, the relationship is governed by the differential accumulation of the potential:

$$\boxed{\frac{d}{dr}(r\,\kappa^2) = \frac{8\pi G}{c^2}\,r^2 \rho_{matter}(r)} \tag{55}$$

Remark 4.9 (Scope of the reproduced sector)**.** *The differential form above reproduces the static tt-component. This is not the boundary of the framework: through phase parametrization the same projections describe any single relational orbit — bound at any eccentricity $e = 2\beta_p^2/\kappa_p^2 - 1$, and unbound (deflection, lensing)– and rotating systems (R.O.M. Kerr without metric,). GR direct parameters translational mapping. Current boundary's: sec:challenges.*

The Vacuum Solution ($\rho_{matter} = 0$). In the vacuum region outside a central mass, the source density vanishes ($\rho_{matter} = 0$). The field equation implies conservation of the projection budget:

$$\frac{d}{dr}(r\kappa^2) = 0 \quad \Longrightarrow \quad r\kappa^2 = \text{const} = R_s.$$

Thus, we recover the potential law of WILL RG:

$$\boxed{\kappa^2 = \frac{R_s}{r}.}$$

Resolution of Roles

1. The Identity $\kappa^2 = \rho/\rho_{\max}$ describes the state of the *field* geometry.
2. The Equation $(r\kappa^2)' \sim \rho_{\text{matter}}$ describes how *matter* generates geometry.
In vacuum, the generator is zero, but the field persists as the algebraic structure $\kappa^2 = R_s/r$.

4.9 No Singularities, No Hidden Regions

WILL Relational Geometry resolves the singularity problem by naturally constraining the valid range of radial distance. Minimal radial distance is kept at $r_{min} = R_s/2$ by the fundamental invariance of light speed limit. Parameters stay finite, and energy remains bounded. Black holes become the boundary of relational difference between energy states.

Distance is not a foundational primitive but relational tension between energy states (3.8) The amplitude bound $\kappa^2 \leq 1$ (static case) and $\kappa^2 \leq 2$ (extremal rotation) caps the radial distance by closure theorem (3.10):

No Singularities

$$r = R_s/\kappa^2 \geq R_s/2$$

$$\beta_{max}^2 = 1 \Longrightarrow \kappa_{max}^2 = 2 \Longrightarrow r_{min} = R_s/\kappa_{max}^2 \Longrightarrow r_{min} = R_s/2 \tag{56}$$

The point $r = 0$ is absent from the admissible relational range.
The same bound enforces:

$$\rho \leq 2\rho_{\max} \quad \text{and} \quad |P| \leq |2P_{\max}| = c^4/(8\pi G r^2) \tag{57}$$

Another interesting hypothetical explanation of this phenomenon comes from Relational Orbital Mechanics. Dynamic Relational Horizon provides new point of view on the nature of Black Holes where precession of the perihelium becomes equivalent to orbital frequency $\Delta\varphi = 2\pi$. The orbit starts to fold on itself and nullify any orbital motion. Its not a two-surface enclosing a region — so there is no interior to be hidden in the first place. However the concept of dynamic horizon remain purely mathematical and requires more research.

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5 Stage IV Consequences and Direct Applications

5.1 Theoretical Ouroboros

Closure

Ontological principle is proven as the inevitable consequence of geometric consistency. Field Equation $\Longleftrightarrow$ Ontological Principle

We have shown that this single Ontological Principle (2.12), through pure geometric reasoning, necessarily leads to an equation which mathematically expresses the very same equivalence we began with. We started with SPACETIME $\equiv$ ENERGY, from which geometry and physical laws are logically derived, and these derived laws then loop back to intrinsically define and limit the very nature of energy and spacetime, proving the self-consistency of the initial idea.

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY.}}$$

The ratio of geometric scales equals the ratio of energy densities.

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho}{\rho_{max}}}$$

$$\boxed{\begin{array}{c}\text{SPACETIME GEOMETRY}\\ \equiv \\ \text{ENERGY DISTRIBUTION}\end{array}}$$

Summary

All physical structure emerges from the single relational equivalence:

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$$

From this, by enforcing geometric self-consistency, one necessarily arrives at the Unified Geometric Field Equation:

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho}{\rho_{\max}}.}$$

This is not an external law but an intrinsic closure relation: geometry and energy are two mutually defining projections of a single entity. It represents the completion of the theoretical Ouroboros — where the principle generates its own mathematical expression and the expression in turn validates the principle.

From a philosophical and epistemological point of view, this can be considered the crown achievement of any theoretical framework - the "Theoretical Ouroboros". But regardless of the aesthetic beauty of this result, let us remain sceptical.

5.2 $W_{\mathbf{ILL}}$: Unity of Relational Structure

The ontological principle

$$\text{SPACETIME} \equiv \text{ENERGY}$$

states that there is only one closed relational resource - WILL. What we call space, time and energy are different projections of the same relational structure. For any energy-closed system observed from a relational origin, this resource appears through four operational projections:

Desmos project

WILL= 1 Derivation of the W_{ILL} invariant.

These quantities are defined as correlated projections of the same underlying WILL structure. In the dimensionful form we write:

$$M = \frac{\beta^2}{\beta_Y}\frac{c^2 a}{G}, \qquad E = \frac{\kappa^2}{\kappa_X}\frac{c^4 a}{2G}, \qquad T = \kappa_X\left(\frac{2Gm_0}{\kappa^2 c^3}\right)^2, \qquad L = \beta_Y\left(\frac{Gm_0}{\beta^2 c^2}\right)^2,$$

where (β, β_Y) and (κ, κ_X) are the kinematic and gravitational projections on the carriers S^1 and S^2, m_0 is the rest mass, $E_0 = m_0 c^2$ is the rest energy, and a is the relational scale of the system as semi-major axis (average length per cycle within this system).

The relational conservation of the carriers $\beta^2 + \beta_Y^2 = 1$, $\quad \kappa_X^2 + \kappa^2 = 1$, together with the closure theorem $\kappa^2 = 2\beta^2$, fix a unique dimensionless combination of these four projections. Combining E, T, M, and L we obtain

$$W_{\text{ILL}} \equiv \frac{ET}{ML} = \frac{\dfrac{E_0}{\kappa_X}\kappa_X t^2}{\dfrac{m_0}{\beta_Y}\beta_Y a^2} = \frac{E_0 t^2}{m_0 a^2}.$$

By the relations that tie temporal $t = a/c$ and spatial $a = R_s/\kappa^2$ scales this ratio is identically equal to unity for any relationally closed system:

$$\boxed{W_{\text{ILL}} = \frac{ET}{ML} = 1.}$$

All dimensionful constants cancel automatically; the value is fixed by the geometry of the carriers, not by a choice of units.

Desmos project

WILL EARTH GPS calculates GPS satellite time shift and testing $W_{ILL} = 1$ on the Earth-GPS system

The same invariant can be written in a phase–normalized form, using local projections

$$E_o = \frac{E_0}{\kappa_{Xo}}, \qquad M_o = \frac{m_0}{\beta_{Yo}}, \qquad T_o = \kappa_{Xo} t_o^2, \qquad L_o = \beta_{Yo} r_o^2,$$

Equivalently, for any state (β_o, κ_o), any scale r_o and any phase along the orbit. Then

$$\boxed{W_{\text{ILL}} = \frac{E_o T_o}{M_o L_o} = 1 \quad \text{for ALL ENERGY'S, ALL SCALES and ALL PHASES.}}$$

$$\frac{E_o}{M_o} = \frac{L_o}{T_o},$$

so the energy sector and the spacetime sector are not independent. Every change of the relational state rescales (E_o, M_o) and (T_o, L_o) coherently so that this equality is always preserved. The familiar practice of treating energy–mass and space–time as separate blocks is therefore an interpretational approximation. This equation does not bring any new predictive power but the ontological meaning shows that any coherent unit system will naturally collapse in to unity dissolving the false separation:

$$\text{SPACETIME} \equiv \text{ENERGY}$$

$$\boxed{W_{ILL} = 1}$$

One conserved relational resource appears as mass, energy, time and length, but always in a way that keeps their ratio fixed.

5.2.1 Interpretive Note: The Name "WILL"

The term **WILL** stands for **SPACE-TIME-ENERGY**. It is both a formal shorthand and a philosophical statement: the universe is not a stage where energy acts through time upon space, but a single self-balancing structure whose internal distinctions generate all phenomena. The name also serves as a gentle irony toward anthropic thinking: the Cosmos does not possess "will" - yet through WILL, it manifests All that Is.

Summary

$$\boxed{\textbf{WILL} \equiv \frac{ET}{ML} = 1 \quad \Longleftrightarrow \quad \text{Geometry} = \text{Energy} = \text{Causality.}}$$

WILL is not the unit of something - but the Unity of Everything.

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5.3 From the 1PN Lagrangian to R.O.M.: A Falsifiable Closed-Form Alternative

Having shown in section "Two-Point Origin of the Lagrangian and Hamiltonian" (4.3) that the Lagrangian and Hamiltonian are artefacts of forcing $r_A = r_B$, we now demonstrate that the relational ontology supplies a complete, non-perturbative replacement. Starting from the standard 1PN Lagrangian only as a reference point, we translate it into the dimensionless projections β, κ and impose the closure (thm:closure) condition (cor:condition) that follows from methodological principles (sec:foundational) of Relational Geometry.

In standard celestial mechanics the relativistic two-body problem is treated by appending post-Newtonian (PN) corrections to the Newtonian Lagrangian or Hamiltonian. The equations of motion are then solved perturbatively, and observables such as the periastron advance emerge as power series in v/c and GM/rc^2. The complexity grows combinatorially with each PN order.

Relational Geometry replaces this entire perturbative machinery by a *finite* set of dimensionless algebraic relations that classify the allowed bound states of a two-body system. These relations are not a truncated series; they are exact consequences of the closure of the underlying relational carriers. Importantly, R.O.M. is *not* a re-summation of the GR PN expansion. It yields a closed-form periastron advance that agrees with GR at first post-Newtonian order but differs at second and higher orders — a deviation that is currently below observational thresholds and thus provides a sharp, falsifiable prediction.

This section gives an explicit, step-by-step mapping from the standard 1PN Lagrangian to the R.O.M. algebraic system and highlights exactly where the two methods coincide and where they diverge.

5.3.1 The standard 1PN test-mass Lagrangian

For a test particle of mass m moving in the field of a central mass M, the Lagrangian in harmonic coordinates, expanded to first post-Newtonian order and divided by m, is

$$L = -c^2 + \frac{1}{2}v^2 + \frac{GM}{r} + \frac{1}{8c^2}v^4 + \frac{3}{2c^2}\frac{GM}{r}v^2 - \frac{1}{2c^2}\left(\frac{GM}{r}\right)^2 + \mathcal{O}(c^{-4}). \tag{58}$$

The terms are, in order: rest energy, Newtonian kinetic and potential energy, the first relativistic correction to kinetic energy, the kinetic-potential coupling, and the non-linear correction to the potential. From this Lagrangian one derives equations of motion and integrates them perturbatively to obtain, for example, the periastron advance

$$\Delta\varphi_{\rm 1PN} = \frac{3\pi R_s}{a(1-e^2)}, \qquad R_s = \frac{2GM}{c^2}.$$

5.3.2 Translation into R.O.M. projections

R.O.M. operates with two dimensionless, frame-invariant projections that are directly measurable from spectroscopic and astrometric data sec:spectroscopic shift without invoking G or M.

$$\beta \equiv \frac{v}{c}, \tag{59}$$

$$\kappa^2 \equiv \frac{R_s}{r}. \tag{60}$$

In terms of these projections, the Newtonian potential term becomes $\dfrac{GM}{r} = \dfrac{1}{2}c^2\kappa^2$. Switching to units $c = 1$ for brevity, the 1PN Lagrangian rewrites as

$$L = \frac{1}{2}\beta^2 + \frac{1}{2}\kappa^2 + \frac{1}{8}\beta^4 + \frac{3}{4}\kappa^2\beta^2 - \frac{1}{8}\kappa^4 + \mathcal{O}(\beta^6). \tag{61}$$

This expression is still a perturbative expansion; the true trajectory must be obtained by varying the Lagrangian and solving differential equations.

5.3.3 The R.O.M. closure condition and why the Lagrangian is no longer required.

The foundational role of **closure theorem** (3.19): For any relationally closed the amplitudes β and κ are not independent but satisfy closure condition (3.21):

$$\boxed{\kappa^2 = 2\beta^2}. \tag{62}$$

This relation follows from the equal distribution of the total conserved relational resource among the independent degrees of freedom of the kinematic carrier S^1 (1 DOF) and the potential carrier S^2 (2 DOF). Closure is not an equation of motion; it is a global structural constraint that every admissible orbit must fulfil.

Once closure is imposed, the Lagrangian is no longer required. The permissible orbital states are completely determined by the algebraic consequences of (5.3.3) together with two further invariants that are already implicit in the Newtonian two-body problem (the conservation of the relational binding energy $\beta^2 = \kappa_o^2 - \beta_o^2$ and of the angular momentum $h = r_o\beta_T c$). The result is a closed system of finite, exact identities.

5.3.4 The R.O.M. algebraic system (exact within the relational framework)

For an elliptical orbit with eccentricity e and semi-major axis a, R.O.M. yields the following closed-form relations (true anomaly O):

- **Keplerian architecture (exact, not just Newtonian):**

$$\frac{R_s}{a} = 2\beta^2 \quad\Longrightarrow\quad a = \frac{R_s}{2\beta^2}, \qquad T \;=\; \frac{2\pi a}{\beta c} = \frac{\pi R_s}{\beta^3 c}. \tag{63}$$

- **Orbital shape (the same ellipse, but with a precessing O):**

$$r(O) \;=\; \frac{a(1-e^2)}{1+e\cos O}. \tag{64}$$

- **Periastron precession per orbit (exact R.O.M. expression):**

$$\Delta\varphi_{\mathrm{ROM}} \;=\; \frac{2\pi\,\tau_Y^2}{1-e^2} \;=\; \frac{2\pi\left(3\beta^2 - 2\beta^4\right)}{1-e^2}. \tag{65}$$

- **Time evolution (the same Kepler integral, with the period from (5.3.4)):**

$$t(O) \;=\; \frac{T}{2\pi}\,\zeta(O), \qquad \zeta(O) = \frac{1}{\sqrt{1-e^2}}\int_0^O \frac{d\theta}{(1+e\cos\theta)^2}. \tag{66}$$

These relations are algebraically exact within R.O.M.; they contain no free parameters and no perturbative approximations. The only departure from Newtonian mechanics is the precession formula (5.3.4).

5.3.5 Connection to the 1PN Lagrangian and the point of departure from GR

Substituting $\kappa^2 = R_s/a$ and the closure condition $\beta^2 = \kappa^2/2$ into (5.3.4) yields the equivalent form

$$\Delta\varphi_{\mathrm{ROM}} \;=\; \frac{3\pi R_s}{a(1-e^2)} \;-\; \frac{\pi R_s^2}{a^2(1-e^2)}. \tag{67}$$

The first term is exactly the GR 1PN advance (5.3.1). Thus, at first post-Newtonian order R.O.M. and GR coincide perfectly—the 1PN Lagrangian is therefore the leading-order content of the R.O.M. algebraic system.

The second term, $-\pi R_s^2/[a^2(1-e^2)]$, is the R.O.M. correction at $\mathcal{O}(\beta^4)$. It is *not* equal to the standard GR 2PN periastron advance; it constitutes a specific, built-in prediction of the relational framework. For Mercury this deviation is -7.3×10^{-7} arcseconds per century, roughly 10^{-8} of the total precession and far below current measurement precision. In stronger fields (e.g., the Galactic Centre S-stars, binary pulsars) the difference will become proportionally larger and will eventually be tested by observations.

Falsifiability

R.O.M. does *not* claim to reproduce the full infinite PN series of GR. Instead, it offers a mathematically transparent alternative: a single, closed-form algebraic expression for the periastron advance that agrees with GR at 1PN but makes a definite, non-adjustable prediction at 2PN and beyond. Future high-precision measurements of periastron precession (e.g., from GRAVITY+ observations of S2, or from binary pulsar timing) will either confirm this prediction or rule out the R.O.M. closure law.

5.3.6 Summary of the mapping

Key takeaway

The 1PN Lagrangian is a truncated series in the small parameters β^2 and κ^2. R.O.M. replaces it with a finite algebraic system that is exact within the relational ontology, contains no free parameters, and matches GR at leading order while making a definite, testable prediction at the next order. The mapping is mathematically transparent and eliminates the need for perturbative expansions altogether.

1PN Lagrangian (harmonic coordinates)	R.O.M. algebraic system
$$L = -c^2 + \frac{1}{2}v^2 + \frac{GM}{r} + \frac{v^4}{8c^2} + \frac{3}{2c^2}\frac{GM}{r}v^2 - \frac{1}{2c^2}\left(\frac{GM}{r}\right)^2 + \dots$$	$$\kappa^2 = \frac{R_s}{r}, \quad \beta = \frac{v}{c}$$ Closure: $\kappa^2 = 2\beta^2$ Orbit: $r(O) = \frac{a(1-e^2)}{1+e\cos O}, \quad a = \frac{R_s}{2\beta^2}$ Precession: $\Delta\varphi_{\mathrm{ROM}} = \frac{2\pi(3\beta^2 - 2\beta^4)}{1-e^2}$ Time: $t(O) = \frac{T}{2\pi}\zeta(O), \quad T = \frac{\pi R_s}{\beta^3 c}$
Connection: The GR 1PN advance is the leading term of $\Delta\varphi_{\mathrm{ROM}}$; the R.O.M. expression adds a specific $\mathcal{O}(\beta^4)$ correction that differs from GR 2PN and provides a falsifiable signature of the relational framework.	

5.4 Methodology comparison of General Relativity (GR) with WILL Relational Geometry (R.O.M.)

Webpage Link

https://willrg.com/methodology/

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5.5 Discussion

All WILL RG results including the success of R.O.M. remains a single unbroken generative chain of derivations from methodological constrains with zero flexibility. Relational Orbital Mechanics is not an independent model. It is the direct operational consequence of applying the Foundational Methodological Principles to bound gravitational systems.

5.5.1 Epistemic and Operational Benefits

Operational Benefits: Absence of Differential Formalism and Acceleration

The most immediate operational consequence of the relational construction is that **no differential equations of motion and no concept of acceleration are required**. In standard orbital mechanics the central acceleration (Newtonian or geodesic) is the primitive from which orbital period, shape, precession, and equilibrium configurations are derived. In WILL RG the same observables are obtained directly from algebraic identities among the projections on the carriers S^1 and S^2, once the methodological principles and the unifying principle SPACETIME $\equiv$ ENERGY have fixed the geometry of those carriers and the unique exchange rate between them.

This absence is not a simplification imposed after the fact; it follows necessarily from the generative structure. Because the theory begins by removing the separation between structure and dynamics, and because it enforces topological closure and maximal symmetry from the outset, the allowed relational states of a bound system are completely classified by the dimensionless projections and the single closure condition that the methodology itself produces. No integration of trajectories, no effective potentials, and no auxiliary constructs such as rotating frames are needed. The mathematics remains a short sequence of explicit algebraic relations whose every step retains clear physical meaning.

The gain in transparency is therefore structural. Observables that in conventional treatments appear only after lengthy perturbative expansions or numerical integration of differential equations emerge here as immediate, exact consequences of the relational geometry. This is visible across all derivations in the paper, including the period–semi-major-axis relation, eccentricity, secular precession, light deflection, and the location of the Sun–Earth L1 point (Sections Relational Parameterization and Sun–Earth L1 point).

5.5.2 Ontological Shift: From Descriptive to Generative Physics

In conventional physics the methodology follows a descriptive paradigm:

1. Observable phenomena are identified.
2. Empirical regularities are codified as "laws of nature."
3. Mathematical formalisms are constructed to *describe* these regularities.

Thus, physical laws are always introduced as external assumptions that model what is seen. Even in General Relativity, where geometry plays the central role, the equivalence principle and the metric postulate are still external inputs.

RG inverts this paradigm. Laws are not added on top of observations; they are *generated* as inevitable consequences of relational geometry:

- There are no independent axioms such as "inertial mass equals gravitational mass."
- Such relations appear automatically as algebraic identities enforced by the geometry.
- What classical physics calls "laws of nature" are secondary shadows of the single relational principle:

$$\text{SPACETIME} \equiv \text{ENERGY}.$$

Summary

Standard Physics: Laws *describe* what we observe.
Relational Geometry: Laws are *generated* as necessary products of closure and self-consistency.

In this sense, the ontological role of physical law is transformed. Physics ceases to be a catalog of empirical descriptions, and becomes the logical unfolding of a single relational structure. WILL identifies the necessary conditions under which all observed phenomena arise.

Descriptive Physics (Standard)	**Generative Physics (WILL)**
Phenomena are *observed* first, then summarized into empirical laws.	Laws emerge as *inevitable consequences* of relational geometry.
Physical laws are *assumptions* introduced to model reality.	Physical laws are *identities*, enforced by geometric self-consistency.
Time and space are treated as external backgrounds.	Time and space are *projections of energy relations.*
Dynamics = evolution of states *in time.*	Dynamics = ordered succession of balanced configurations; *time is emergent.*
Goal: *describe* what is observed.	Goal: *show why nothing else is possible.*

Table 5: Ontological contrast between standard descriptive physics and the generative paradigm of WILL Relational Geometry.

Phenomenon	Standard (GR) Result	Relational Geometry (RG)
GPS time shift / gravitational redshift	Frequency shift = combination of kinetic (SR) and gravitational (GR) effects.	Single symmetric law: $\tau = \beta_Y \cdot \kappa_X$, $E_{\text{loc}} = \frac{E_0}{\sqrt{(1-\beta^2)(1-\kappa^2)}} = \frac{E_{\text{loc}}}{\tau}$ 38.52 μs/day verified directly with GPS satellites.
Photon sphere, ISCO, horizons	Derived by solving geodesic equations in Schwarzschild metric.	Critical radii emerge from simple symmetry's (Photon sphere: $\theta_1 = \theta_2 = 54.73°$ ("magic angle") $Q^2 = \kappa^2 + \beta^2 = 1$. ISCO: $Q = Q_Y$,
Mercury's perihelion precession	Complex expansion of Einstein field equations.	Same number obtained from RG with $\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1-e^2} = 43''$/century. Using simple algebra.
Cosmological redshift	Photon "loses energy" as universe expands.	Energy conserved; redshift = redistribution of projection parameters. (Details in WILL PART II)
Cosmological absolute scale (Supernovae fit)	Hubble-like expansion, ΛCDM fits	$H_0 \equiv \sqrt{8\pi G\rho_\gamma/(3\alpha^2)} = 68.15$ Derived from CMB temperature and α connecting micro and macro scales (Details in WILL PART II-III).
Cosmological constant Λ	Added by hand to fit data ("dark energy").	Arises naturally as $\Lambda = 2/3r^2$. No extra entities required. (More details in WILL PART I and II)
Singularities	Predicted in black holes and big bang ($\rho \to \infty$).	Forbidden: density bounded by $\rho_{\max} = c^2/(8\pi G r^2)$.
Local gravitational energy	"Cannot be localized" (only ADM/Bondi at infinity).	Directly measurable via κ, e.g. from light deflection angle or red shift.
Unification with QM	No natural unification in GR framework.	Same projectional law applies from microscopic $\alpha = \beta_1$ (QM) to cosmic $\kappa^2 = \Omega_\Lambda$ (GR, COSMO) scales. (Details in WILL PART II and III)

Table 6: Classical GR results vs. WILL RG outcomes. Known effects are recovered by simpler symmetric laws, while new predictions eliminate singularities and explain cosmology without dark entities.

5.6 Challenges and Current Limitations

Gravitational waves and the full radiative two-body problem have not yet been formulated within the relational framework. The current construction addresses only conservative, bound orbits. Extending it to dissipative, radiative dynamics requires a relational description of propagating degrees of freedom on the carriers and remains an open technical task.

A complete, self-consistent multi-body architecture that extends beyond the collinear L1 case has likewise not been constructed. The nested-contour approach provides a clear conceptual route, but its explicit algebraic realisation for systems containing three or more gravitationally bound bodies (including mean-motion resonances and long-term stability) requires further development. The author states openly that progress on both fronts has so far been limited by present mathematical facility rather than by any intrinsic prohibition within the relational ontology. Collaboration with researchers experienced in radiative dynamics and in the algebraic treatment of multi-body closures is therefore actively invited.

5.7 Falsifiability and Future Tests

The framework yields several sharp, near-term testable predictions that follow directly from its algebraic structure.

The most immediate discriminator is continued high-precision monitoring of the S2 star. The relational model predicts that the true-anomaly-coupled precession parametrisation will preserve or strengthen its statistical advantage over the conventional time-linear implementation as new pericentre passages are added with improved GRAVITY+ astrometry (2026–2028 and subsequent windows). A statistically significant reversal of the present $\Delta\chi^2$ improvement on the enlarged dataset would falsify the relational phasing hypothesis while remaining compatible with the magnitude of the precession itself.

For Solar-System tests, the explicit second-order correction in the RG. precession formula is in principle accessible with sufficiently long observational baselines or with planets at smaller semi-major axes. A confirmed detection at the predicted magnitude, or a firm upper bound below it, would directly constrain the cross-coupling term that is absent from the first-order post-Newtonian expression.

These tests are decisive because they probe either the phasing of precession or the explicit higher-order content of the closure theorem – both of which are uniquely fixed by the relational algebra and are not adjustable parameters.

5.8 Conclusion

Relational Geometry demonstrates that the entire phenomenology of motion – from Keplerian architecture to perihelion precession and light deflection – emerges as an inevitable algebraic consequence of the Foundational Methodological Principles. All WILL RG results including the complete success of R.O.M. remains an unbroken chain of derivations from firs principles under extreme methodological constrains with zero flexibility.

This is not a re-description of General Relativity; it is the irreducible algebraic core that remains when the ontological superstructure of Riemannian geometry and tensorial formalism is removed. The S2 orbit fit and Mercury's precession clearly show that RG is not the subset of GR but separate model with radically cleaner and lighter ontological baggage. Future more accurate astronomical data will be able to definitively show which ontology the empirical data gives preference to.

Within RG several structural consequences follow:

1. **G and M are secondary.** The Schwarzschild length R_s and all orbital distances are derived directly from dimensionless spectroscopic and chronometric ratios. The Newtonian gravitational constant and the kilogram mass appear only as conversion factors when translating geometric results into legacy units.

2. **Derivation of Metrics from Relational Geometry.** The Minkowski, Schwarzschild and Kerr metrics and there intervals — revealed not as fundamental postulates of a 4D manifold, but as the Pythagorean identities of the relational carriers S^1 and S^2. Relativistic effects—such as time dilation and length contraction—emerge naturally as geometric rotations between external interaction (Amplitude) and internal existence (Phase).

3. **Closure replaces the virial theorem.** The relation $\kappa^2 = 2\beta^2$ is not an empirical input but the inevitable consequence of equitable budget distribution across the independent degrees of freedom of the kinematic and potential carriers.

4. **Precession is topological, not dynamical.** The secular advance of the pericentre arises from the mismatch of internal phase over one cyclic extension of the orbital phase, without integrating geodesic equations or expanding in powers of v/c.

5. **Singularity's are avoided** by naturally constraining the valid range of radial distance. **The point $r = 0$ is absent from the admissible relational range.**

6. WILL trilogy offers a unified, parameter-free ground-up unbroken chain derivation of most major observed phenomena within extreme methodological constrains and 0 flexibility. The traditional unjustified separation of spacetime and energy dissolves into a single generative structure we call WILL.

The results presented here invite exacting numerical tests (additional binary pulsars, triple systems, high-resolution S2 pericentre passages) and provide an operational language in which gravitational phenomena can be communicated without surplus ontology. The hope is that this physical and mathematical clarity returned by RG's methodology will help turn the focus of physical science back to its earliest ambition: to understand the logic of the world, not just to model it.

Final Summary

SPACETIME $\equiv$ ENERGY.

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